

# Supplementary Material for: Martingale Innovations from Recurrent Networks for Sequential Change-Point Detection

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## A. Background and Preliminaries

We work throughout on a complete probability space  $(\Omega, \mathcal{A}, \mathbb{P})$ . All convergence statements are in the almost-sure or  $L^p$  sense unless otherwise noted.

### A.1. Filtrations and the Information Flow

**Definition A.1 (Filtration and natural filtration).** A *filtration* is an increasing family of sub- $\sigma$ -algebras  $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{A}$ . For a sequence  $(X_t)_{t \geq 1}$  of random variables, the *natural filtration* is  $\mathcal{F}_t = \sigma(X_1, \dots, X_t)$  and its completion  $\mathcal{F}_\infty^+ = \sigma(\bigcup_{t \geq 1} \mathcal{F}_t)$ .

A *decreasing filtration* (or *reverse filtration*) is a family  $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \dots$  of sub- $\sigma$ -algebras. The canonical example relevant to this paper is

$$\mathcal{G}_n = \sigma(X_n, X_{n+1}, \dots), \quad (\text{A.1})$$

which formalizes the information contained in the *tail* of the sequence starting at index  $n$ .

**Definition A.2 (Tail  $\sigma$ -algebra).** The *tail  $\sigma$ -algebra* of a sequence  $(X_t)_{t \geq 1}$  is

$$\mathcal{T}_\infty = \bigcap_{n=1}^{\infty} \sigma(X_n, X_{n+1}, \dots) = \bigcap_{n=1}^{\infty} \mathcal{G}_n.$$

An event  $A \in \mathcal{T}_\infty$  is *tail-measurable*: its occurrence or non-occurrence is not affected by any finite initial segment of the sequence.

**Example A.3 (Tail-measurable functionals).** For an i.i.d. or ergodic sequence  $(X_t)$ : (a) the limit  $\lim_{n \rightarrow \infty} n^{-1} \sum_{t=1}^n X_t$  (when it exists) is  $\mathcal{T}_\infty$ -measurable; (b) the event  $\{\limsup_{t \rightarrow \infty} X_t > c\}$  is tail-measurable; (c) the population mean  $\mu = \mathbb{E}[X_1]$  is a constant, hence trivially  $\mathcal{T}_\infty$ -measurable under any probability measure. Finite-dimensional functions of  $(X_1, \dots, X_k)$  for fixed  $k$  are *not* tail-measurable in general.

### A.2. Martingales and Reverse Martingales

**Definition A.4 (Martingale and martingale difference sequence).** A sequence  $(M_t, \mathcal{F}_t)_{t \geq 1}$  is a *martingale* if  $\mathbb{E}[|M_t|] < \infty$  and  $\mathbb{E}[M_{t+1} | \mathcal{F}_t] = M_t$  a.s. for all  $t$ . The sequence  $(D_t)_{t \geq 1}$  is a *martingale difference sequence* (MDS) if  $\mathbb{E}[|D_t|] < \infty$  and  $\mathbb{E}[D_t | \mathcal{F}_{t-1}] = 0$  a.s. for all  $t \geq 2$ .

The posterior mean  $\mathbb{E}[\theta \mid \mathcal{F}_t]$ , for a fixed integrable parameter  $\theta$  and observations  $X_1, X_2, \dots$ , is the canonical martingale in Bayesian sequential analysis.

**Definition A.5 (Reverse martingale).** Let  $(\mathcal{G}_n)_{n \geq 1}$  be a decreasing filtration. A sequence  $(Y_n, \mathcal{G}_n)_{n \geq 1}$  is a *reverse martingale* if  $\mathbb{E}[|Y_n|] < \infty$  and

$$\mathbb{E}[Y_n \mid \mathcal{G}_{n+1}] = Y_{n+1} \quad \text{a.s. for all } n \geq 1.$$

A prototypical reverse martingale is obtained by taking an integrable random variable  $Y$  and setting  $Y_n = \mathbb{E}[Y \mid \mathcal{G}_n]$  for a decreasing filtration  $(\mathcal{G}_n)$ . Exchangeability arguments also represent empirical averages through appropriate decreasing sigma-fields. The key convergence theorem is:

**Theorem A.6 (Reverse martingale convergence; Doob 1953).** If  $(Y_n, \mathcal{G}_n)_{n \geq 1}$  is an  $L^1$ -bounded reverse martingale with decreasing filtration  $(\mathcal{G}_n)$ , then

$$Y_n \xrightarrow{\text{a.s. and } L^1} Y_\infty := \mathbb{E}[Y_1 \mid \mathcal{G}_\infty], \quad \mathcal{G}_\infty = \bigcap_{n \geq 1} \mathcal{G}_n.$$

**Proof.** Let  $(Y_n, \mathcal{G}_n)_{n \geq 1}$  be an  $L^1$ -bounded reverse martingale with decreasing filtration  $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \dots$  and  $\mathcal{G}_\infty = \bigcap_{n \geq 1} \mathcal{G}_n$ . Without loss of generality, write  $Y_n = \mathbb{E}[Y_1 \mid \mathcal{G}_n]$ , which is consistent with the reverse martingale property by the tower law: for  $m > n$ ,  $\mathcal{G}_m \subseteq \mathcal{G}_n$  implies  $\mathbb{E}[Y_n \mid \mathcal{G}_m] = \mathbb{E}[\mathbb{E}[Y_1 \mid \mathcal{G}_n] \mid \mathcal{G}_m] = \mathbb{E}[Y_1 \mid \mathcal{G}_m] = Y_m$  a.s.

**Almost-sure convergence.** For any  $a < b$ , let  $U_n([a, b])$  be the number of upcrossings of  $[a, b]$  by the sequence  $Y_1, Y_2, \dots, Y_n$ . Doob's upcrossing inequality for reverse martingales (Neveu, 1975, Chapter V) gives

$$(b - a) \mathbb{E}[U_n([a, b])] \leq \mathbb{E}[(Y_1 - a)^+] \leq \mathbb{E}[|Y_1|] + |a| < \infty.$$

Letting  $n \rightarrow \infty$ , the monotone convergence theorem gives  $\mathbb{E}[U_\infty([a, b])] < \infty$ , so  $U_\infty([a, b]) < \infty$  a.s. Since  $a < b$  are arbitrary rationals, the number of upcrossings of every rational interval is finite almost surely, which implies  $Y_n \rightarrow Y_\infty$  a.s. for some  $[-\infty, \infty]$ -valued random variable  $Y_\infty$ .

**$L^1$  convergence and integrability of the limit.** For each  $n$ ,  $|Y_n| = |\mathbb{E}[Y_1 \mid \mathcal{G}_n]| \leq \mathbb{E}[|Y_1| \mid \mathcal{G}_n]$  by Jensen's inequality. The family  $\{\mathbb{E}[|Y_1| \mid \mathcal{G}_n]\}_{n \geq 1}$  is uniformly integrable by the standard fact that the collection of conditional expectations  $\{\mathbb{E}[Z \mid \mathcal{G}] : \mathcal{G} \subseteq \mathcal{A}\}$  of a fixed  $Z \in L^1$  is uniformly integrable (Williams, 1991, Sec. 13.4). Hence  $\{Y_n\}$  is uniformly integrable. Uniform integrability together with a.s. convergence implies  $L^1$  convergence:  $\|Y_n - Y_\infty\|_{L^1} \rightarrow 0$ , and in particular  $Y_\infty \in L^1$ .

**Identification of the limit.** For any  $A \in \mathcal{G}_\infty \subseteq \mathcal{G}_n$ , the definition of conditional expectation gives  $\int_A Y_n d\mathbb{P} = \int_A Y_1 d\mathbb{P}$  for all  $n$ . Passing to the limit using  $L^1$  convergence:  $\int_A Y_\infty d\mathbb{P} = \int_A Y_1 d\mathbb{P}$  for all  $A \in \mathcal{G}_\infty$ . By the definition of conditional expectation,  $Y_\infty = \mathbb{E}[Y_1 \mid \mathcal{G}_\infty]$  a.s.  $\square$

**Remark A.7 (Comparison with forward convergence).** Doob's *forward* martingale convergence theorem requires  $L^1$ -boundedness and yields almost-sure convergence but not necessarily  $L^1$  convergence (unless the martingale is uniformly integrable). Reverse martingales are automatically uniformly integrable, so both modes of convergence are free. This property is useful for tail-projection arguments such as Lemma 2.1 of the main paper.

### A.3. The Hewitt–Savage Zero-One Law

**Theorem A.8 (Hewitt–Savage; Hewitt and Savage 1955).** *Let  $(X_t)_{t \geq 1}$  be an i.i.d. sequence. Then every event invariant under finite permutations of the coordinates has probability 0 or 1. In particular, every tail event  $A \in \mathcal{T}_\infty$  satisfies  $\mathbb{P}(A) \in \{0, 1\}$ .*

For i.i.d. sequences, Theorem A.8 implies that the tail  $\sigma$ -algebra is trivial, so  $\mathbb{E}[Z \mid \mathcal{T}_\infty] = \mathbb{E}[Z]$  for any integrable  $Z$ . The statement should not be extended to arbitrary exchangeable sequences: by de Finetti’s theorem, an exchangeable but non-i.i.d. Bernoulli sequence can have a nontrivial tail generated by the random mixing parameter. For stationary ergodic sequences, the ergodic theorem identifies limits of time averages, but it does not imply that every tail-measurable functional is literally a time average.

### A.4. Recurrent Neural Networks and LSTMs

We give precise definitions of the two architectures whose hidden states we analyze.

**Definition A.9 (Recurrent Neural Network).** An RNN processes a sequence  $(x_t)_{t=1}^T$ ,  $x_t \in \mathbb{R}^p$ , via

$$h_t = \phi(W_h h_{t-1} + W_x x_t + b_h), \quad (\text{A.2})$$

where  $h_t \in \mathbb{R}^d$  is the *hidden state*,  $W_h \in \mathbb{R}^{d \times d}$ ,  $W_x \in \mathbb{R}^{d \times p}$ ,  $b_h \in \mathbb{R}^d$ , and  $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a coordinate-wise non-linear activation (e.g. tanh or ReLU). The initial state  $h_0 \in \mathbb{R}^d$  is fixed or learnable; the  $L^2$  theory below assumes it is square-integrable.

**Remark A.10 (Notation).** In this appendix  $d$  denotes the hidden-state dimension and  $p$  the input dimension. In the main paper (Section 2), the same dimensions are written  $p$  (hidden) and  $d_{\text{in}}$  (input); the symbol  $d$  in the main paper refers to the observation dimension in the empirical studies (Section 4).

**Assumption A.11 (Effective operator-norm contractivity, restated).** This restates the main paper’s contractivity assumption for the reader’s convenience; it is not a separate hypothesis. Throughout the paper we assume

$$L_\phi \|W_h\|_{\text{op}} < 1,$$

where  $L_\phi$  is the Lipschitz constant of the activation  $\phi$  and  $\|W_h\|_{\text{op}} = \sigma_{\max}(W_h)$  is the operator norm. This is the contraction needed for  $L^2$ -stability of (A.2). The spectral radius alone is insufficient for non-normal matrices because it does not control the vector norm. Spectral normalization can encourage or enforce this condition after accounting for  $L_\phi$  (Miyato et al., 2018).

**Definition A.12 (Long Short-Term Memory).** An LSTM (Hochreiter and Schmidhuber, 1997) augments the basic RNN with a cell state  $c_t \in \mathbb{R}^d$  and three gating mechanisms:

$$f_t = \sigma(W_f[h_{t-1}; x_t] + b_f), \quad (\text{A.3})$$

$$i_t = \sigma(W_i[h_{t-1}; x_t] + b_i), \quad (\text{A.4})$$

$$o_t = \sigma(W_o[h_{t-1}; x_t] + b_o), \quad (\text{A.5})$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}; x_t] + b_c), \quad (\text{A.6})$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (\text{A.7})$$

$$h_t = o_t \odot \tanh(c_t), \quad (\text{A.8})$$

where  $[h_{t-1}; x_t] \in \mathbb{R}^{d+p}$  denotes concatenation,  $\sigma$  is the logistic sigmoid, and  $\odot$  is coordinate-wise multiplication. The *forget gate*  $f_t \in (0, 1)^d$  controls how much of the previous cell state survives to time  $t$ .

**Remark A.13 (Contractivity of LSTM).** The coordinate-wise bound  $f_t \in (0, 1)^d$  does not by itself make the full LSTM cell recursion uniformly contractive. The gate can be arbitrarily close to one, and  $f_t$  depends on  $h_{t-1}$ , which depends on  $c_{t-1}$ , so derivatives of the gate map also enter the Lipschitz constant. A rigorous contraction theorem for LSTMs therefore requires a separate assumption, such as a uniform gate bound  $\|f_t\|_\infty \leq \gamma < 1$  together with Lipschitz control of the gate maps. In this paper, the rigorous contraction theorem is stated only for the basic RNN recurrence.

## A.5. Stationary Ergodic Processes

**Definition A.14 (Stationarity and ergodicity).** A process  $(X_t)_{t \geq 1}$  is (*strictly*) *stationary* if

$$(X_{t+k})_{t \geq 1} \stackrel{d}{=} (X_t)_{t \geq 1}$$

for all  $k \geq 1$ . It is *ergodic* if every shift-invariant event has probability 0 or 1. By the ergodic theorem (Walters, 1982), for any  $L^1$  functional  $g$ ,

$$\frac{1}{T} \sum_{t=1}^T g(X_t) \xrightarrow{a.s.} \mathbb{E}[g(X_1)] \quad \text{as } T \rightarrow \infty.$$

Stationarity is the maintained assumption throughout the theory sections (Sections 2–3) of the main paper; the empirical Section 4 deliberately includes non-stationary change-point and real-data settings. The non-stationary extension via locally stationary processes (Dahlhaus, 1997) is treated in Section 2.3 of the main paper.

## B. Proofs

For reference, Table B.1 maps each result of the main paper to the object it concerns, the assumptions it requires, and what it establishes; the proofs follow in order.

### B.1. Proof of Lemma 2.1 of the main paper (Equivalence of Forward and Tail Limits)

**Proof of Lemma 2.1 in the main paper. Part (a).** Since  $\theta \in L^\infty \subseteq L^1$  and  $\theta$  is  $\mathcal{T}_\infty$ -measurable, it is also  $\mathcal{F}_\infty^+$ -measurable because  $\mathcal{T}_\infty \subseteq \mathcal{G}_1 = \sigma(X_1, X_2, \dots) \subseteq \mathcal{F}_\infty^+$ . By Doob’s  $L^1$  martingale convergence theorem (Doob, 1953),  $\mathbb{E}[\theta \mid \mathcal{F}_t] \rightarrow \mathbb{E}[\theta \mid \mathcal{F}_\infty^+]$  a.s. and in  $L^1$ . Since  $\theta$  is  $\mathcal{F}_\infty^+$ -measurable,  $\mathbb{E}[\theta \mid \mathcal{F}_\infty^+] = \theta$  a.s.

**Part (b).** The sequence  $(\mathbb{E}[\theta \mid \mathcal{G}_n], \mathcal{G}_n)_{n \geq 1}$  is a reverse martingale: for  $m > n$ ,  $\mathcal{G}_m \subseteq \mathcal{G}_n$  implies  $\mathbb{E}[\mathbb{E}[\theta \mid \mathcal{G}_n] \mid \mathcal{G}_m] = \mathbb{E}[\theta \mid \mathcal{G}_m]$  by the tower property. Boundedness of  $\theta$  gives  $L^1$ -boundedness. By the reverse martingale convergence theorem (Theorem A.6; Doob 1953),  $\mathbb{E}[\theta \mid \mathcal{G}_n] \rightarrow \mathbb{E}[\theta \mid \mathcal{T}_\infty]$  a.s. and in  $L^1$ . Since  $\theta$  is  $\mathcal{T}_\infty$ -measurable,  $\mathbb{E}[\theta \mid \mathcal{T}_\infty] = \theta$  a.s.

**Part (c).** Immediate from (a) and (b): both limits equal  $\theta$  a.s.

**Table B.1.** Map of the main-paper results. The top block (tier one) concerns the contractive RNN and its innovations; the bottom block (tier two) concerns the Innovation CUSUM detector and requires no RNN contraction theorem, only conditional residual-drift and tail conditions on fitted residuals.

| Result    | Object                           | Key assumptions                                                                                                                       | Establishes                                                                |
|-----------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Thm 2.5   | Contractive state $h_t$          | RNN stationary ergodic input, $\mathbb{E} \ X\ ^2 < \infty$ , $L_\phi \ W_h\ _{\text{op}} < 1$                                        | Doob/MDS decomposition; geometric innovation stabilization                 |
| Cor 2.8   | RNN innovations $M_t$            | additionally $(2+\delta)$ -moment, $\Sigma_M^* \succ 0$                                                                               | functional CLT for cumulative innovations                                  |
| Prop 2.9  | Prediction residuals $e_t$       | Bayes predictor (exact); $L^2$ -accurate fitted predictor (approx.)                                                                   | exact / approximate MDS                                                    |
| Prop 2.10 | Recurrent predictor residuals    | hypotheses of Thm 2.5; Lipschitz readout                                                                                              | residual drift $\leq L_o \rho^t \Delta_0$ + stationary readout error       |
| Prop 2.11 | Fitted vs. linear residuals      | nonlinear conditional mean; consistent learned predictor                                                                              | irreducible linear defect; learned defect $\rightarrow 0$                  |
| Prop 2.13 | Fitted-predictor defect $\eta_T$ | approximation rate $W^{-2\beta}$ ; $\beta$ -mixing excess-risk rate                                                                   | $\eta_T = \tilde{O}(T^{-\beta/(2\beta+1)})$ ; margin $\kappa_T \uparrow k$ |
| Thm 3.2   | Innovation CUSUM score           | drift margin $k > \eta$ , conditional sub-Gaussianity                                                                                 | exponential in-control ARL lower bound                                     |
| Thm 3.6   | Innovation CUSUM score           | post-change drift $\geq k + \mu$ , bounded increments                                                                                 | out-of-control detection-delay upper bound                                 |
| Cor 3.7   | Innovation CUSUM score           | Assumptions above                                                                                                                     | exponential-ARL / logarithmic-delay trade-off                              |
| Thm 3.9   | Innovation CUSUM score           | Assumptions above; Gaussian innovation shift for exactness                                                                            | Lorden order-optimality; exact minimax optimality (Gaussian)               |
| Cor 3.11  | Mahalanobis innovation score     | uniform-in- $d$ : bounded $\eta_d$ , $\lambda_{\min}(\Sigma_0) \geq c_0$ , conditional homoskedasticity, $O(1)$ score sub-Gaussianity | dimension-stable ARL                                                       |

**Parts (d) and (e).** For general  $\theta \in L^1$ , part (a) extends to  $\mathbb{E}[\theta | \mathcal{F}_t] \rightarrow \mathbb{E}[\theta | \mathcal{F}_\infty^+]$  by the same Doob argument. For part (e), the reverse martingale convergence in (b) applies unchanged since the reverse martingale is  $(\mathbb{E}[\theta | \mathcal{G}_n], \mathcal{G}_n)$ , and the limit is  $\mathbb{E}[\theta | \mathcal{T}_\infty]$  regardless of whether  $\theta$  is tail-measurable. The two limits agree if and only if  $\mathbb{E}[\theta | \mathcal{F}_\infty^+]$  is  $\mathcal{T}_\infty$ -measurable, which holds when  $\theta$  itself is tail-measurable.  $\square$

## B.2. Proof of Theorem 2.5 of the main paper (MDS Decomposition with Stabilization Rate)

**Proof of Theorem 2.5 in the main paper.** Throughout,  $\rho = L_\phi \|W_h\|_{\text{op}} < 1$  and  $\Delta_0 = \|h_0 - H_0^*\|_{L^2}$ .

**Part (i):  $L^2$ -boundedness.** We prove the  $C_0$  formula for the case  $\phi(0) = 0$  (e.g. tanh, ReLU), where the Lipschitz bound gives  $\|\phi(u)\| \leq L_\phi \|u\|$ . (For activations with  $\phi(0) \neq 0$ , e.g. sigmoid, an additional term  $L_\phi \|\phi(0)\| / (1 - \rho)$  is added to  $C_0$ , consistent with the general formula stated in Theorem 2.5 of the main paper.) By the Lipschitz bound  $\|\phi(u)\| \leq L_\phi \|u\|$  and Minkowski's inequality in  $L^2$ ,

$$\|h_t\|_{L^2} \leq L_\phi \|W_h h_{t-1} + W_x X_t + b_h\|_{L^2} \leq \rho \|h_{t-1}\|_{L^2} + L_\phi \|W_x\|_{\text{op}} \|X_t\|_{L^2} + L_\phi \|b_h\|.$$

Let  $c := L_\phi(\|W_x\|_{\text{op}}\|X_1\|_{L^2} + \|b_h\|)$ , which is finite by stationarity. Iterating the inequality and summing the geometric series gives

$$\|h_t\|_{L^2} \leq \rho^t \|h_0\|_{L^2} + \frac{c}{1-\rho},$$

so  $\sup_t \|h_t\|_{L^2} \leq \|h_0\|_{L^2} + c/(1-\rho) =: C_0 < \infty$ .

**Part (ii): MDS structure.** Since  $h_t$  is  $\mathcal{F}_t$ -measurable and square-integrable by part (i), the conditional expectation  $D_t := \mathbb{E}[h_t | \mathcal{F}_{t-1}]$  is well-defined and  $\mathcal{F}_{t-1}$ -measurable. Setting  $M_t := h_t - D_t$ ,

$$\mathbb{E}[M_t | \mathcal{F}_{t-1}] = \mathbb{E}[h_t - D_t | \mathcal{F}_{t-1}] = D_t - D_t = 0 \quad \text{a.s.}$$

**Part (iii): Innovation  $L^2$  bound.** Conditional on  $\mathcal{F}_{t-1}$ , the previous state  $h_{t-1}$  is fixed and the only new random input is  $X_t$ . Put  $\mu_t = \mathbb{E}[X_t | \mathcal{F}_{t-1}]$  and define  $g_t(x) := \phi(W_h h_{t-1} + W_x x + b_h)$ . Since  $h_{t-1}$  is  $\mathcal{F}_{t-1}$ -measurable,  $g_t(\mu_t) = \phi(W_h h_{t-1} + W_x \mu_t + b_h)$  is  $\mathcal{F}_{t-1}$ -measurable. By the best-approximation property of the conditional expectation ( $D_t = \mathbb{E}[h_t | \mathcal{F}_{t-1}]$  minimizes the  $L^2$  distance to  $h_t$  over all  $\mathcal{F}_{t-1}$ -measurable functions),

$$\begin{aligned} \mathbb{E}\{\|M_t\|^2 | \mathcal{F}_{t-1}\} &\leq \mathbb{E}\{\|g_t(X_t) - g_t(\mu_t)\|^2 | \mathcal{F}_{t-1}\} \\ &\leq L_\phi^2 \|W_x\|_{\text{op}}^2 \mathbb{E}\{\|X_t - \mu_t\|^2 | \mathcal{F}_{t-1}\}. \end{aligned}$$

Taking unconditional expectations: since  $\mathbb{E}\{\|X_t - \mu_t\|^2 | \mathcal{F}_{t-1}\} = \text{Var}(X_t | \mathcal{F}_{t-1})$  is the conditional variance, the law of total variance gives  $\mathbb{E}[\text{Var}(X_t | \mathcal{F}_{t-1})] = \text{Var}_2(X_t) - \mathbb{E}\|\mu_t - \mathbb{E}X_t\|^2 \leq \text{Var}_2(X_t)$ , which yields equation (1) in the main paper.

**Part (iv): Innovation stabilization rate.** Write  $M_t - M_t^* = (h_t - H_t^*) - (D_t - D_t^*)$ . By the triangle inequality,  $\|M_t - M_t^*\|_{L^2} \leq \|h_t - H_t^*\|_{L^2} + \|D_t - D_t^*\|_{L^2}$ . The coupling bound (Proposition B.1 below) gives  $\|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0$ . Since conditional expectation is an  $L^2$  contraction,  $\|D_t - D_t^*\|_{L^2} = \|\mathbb{E}[h_t - H_t^* | \mathcal{F}_{t-1}]\|_{L^2} \leq \|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0$ . Adding the two bounds gives  $\|M_t - M_t^*\|_{L^2} \leq 2\rho^t \Delta_0$ .

**Part (v): Variance decomposition and  $\Sigma_M^*$ .** The law of total variance applied coordinatewise, combined with  $\mathbb{E}[M_t | \mathcal{F}_{t-1}] = 0$ , gives

$$\mathbb{E}\|h_t - \mathbb{E}h_t\|^2 = \mathbb{E}\|D_t - \mathbb{E}h_t\|^2 + \mathbb{E}\|M_t\|^2,$$

since the cross term  $2\mathbb{E}\langle D_t - \mathbb{E}h_t, M_t \rangle = 0$ .

For the stationary regime, recall that the filtration is the two-sided past  $\mathcal{F}_{t-1} = \sigma(X_s : s \leq t-1) \vee \sigma(h_0)$  with  $h_0$  independent of the process (main paper, Section 2 setup). By the causal construction (Proposition B.1),  $H_t^*$  is the same fixed measurable functional of  $(\dots, X_{t-1}, X_t)$  at every  $t$ , and since  $\sigma(h_0)$  is independent of  $\sigma(X_s : s \in \mathbb{Z})$ , conditioning on it leaves the conditional expectation unchanged:  $D_t^* = \mathbb{E}[H_t^* | \mathcal{F}_{t-1}] = \mathbb{E}[H_t^* | \sigma(X_s : s \leq t-1)]$  is likewise the same functional of the shifted path at every  $t$ . Hence  $(M_t^* = H_t^* - D_t^*)_{t \in \mathbb{Z}}$  is strictly stationary and  $\Sigma_M^* := \mathbb{E}[M_t^* (M_t^*)^\top]$  is independent of  $t$ . (With the one-sided filtration  $\sigma(X_1, \dots, X_{t-1})$  this would fail: the conditioning set is not shift-invariant.) Positive semi-definiteness is immediate. The bound  $\Sigma_M^* \preceq \sigma_M^2 I_p$  follows from part (iii) applied to the stationary solution: for a positive semi-definite matrix,  $\lambda_{\max}(\Sigma_M^*) \leq \text{tr}(\Sigma_M^*) = \mathbb{E}\|M_t^*\|^2 \leq \sigma_M^2$ .

For strict positivity of  $\Sigma_M^*$ : suppose for contradiction that  $\lambda_{\min}(\Sigma_M^*) = 0$ . Then there exists a unit vector  $v \in \mathbb{R}^p$  with  $\mathbb{E}[(v^\top M_t^*)^2] = 0$ , so  $v^\top M_t^* = 0$  a.s., meaning  $v^\top H_t^*$  is  $\mathcal{F}_{t-1}$ -measurable. But then  $\text{Var}(v^\top H_t^* | \mathcal{F}_{t-1}) = 0$  a.s., which contradicts the nondegeneracy condition (5) in the main paper (equation (5):  $\text{Var}(v^\top \phi(W_h h_{t-1}^* + W_x X_t + b_h) | \mathcal{F}_{t-1}) > 0$  a.s. for all nonzero  $v$ ). Hence  $\lambda_{\min}(\Sigma_M^*) > 0$  under that condition.

*Remark on the full row rank sufficient condition.* A classical sufficient condition for equation (5) is  $\text{rank}(W_x) = p$  together with  $X_t$  not entirely predictable from  $\mathcal{F}_{t-1}$ . When  $W_x$  has full row rank,  $W_x^\top v \neq 0$  for all nonzero  $v$ , which implies  $\text{Var}(v^\top H_t^* | \mathcal{F}_{t-1}) > 0$  a.s. for all nonzero  $v$  under *linear* activation (so that variance propagates linearly through  $g_t$ ). This implication can fail for nonlinear  $\phi$  in a nearly flat or saturating region; the nondegeneracy condition (5) is the correct sufficient condition in general.  $\square$

### B.3. Proof of Corollary 2.8 of the main paper (Functional CLT for MDS Innovations)

**Proof of Corollary 2.8 in the main paper. CLT for the stationary process  $(M_t^*)$ .** Under the two-sided filtration  $\mathcal{F}_t = \sigma(X_s : s \leq t) \vee \sigma(h_0)$  of the main-paper setup,  $H_t^*$  — and hence  $M_t^* = H_t^* - \mathbb{E}[H_t^* | \mathcal{F}_{t-1}]$  — is  $\mathcal{F}_t$ -measurable by the causal construction, so  $(M_t^*, \mathcal{F}_t)_{t \geq 1}$  is an *adapted*, strictly stationary MDS (stationarity by the proof of Theorem 2.5(v) above); the martingale FCLT below requires this adaptedness. We verify  $\sup_t \mathbb{E} \|M_t^*\|^{2+\delta} < \infty$ . By the Lipschitz recursion and Minkowski's inequality in  $L^{2+\delta}$ ,

$$\|H_t^*\|_{L^{2+\delta}} \leq \rho \|H_{t-1}^*\|_{L^{2+\delta}} + L_\phi \|W_x\|_{\text{op}} \|X_t\|_{L^{2+\delta}} + L_\phi \|b_h\|.$$

Stationarity gives a common value  $C = c'/(1 - \rho) < \infty$ , so  $\sup_t \|H_t^*\|_{L^{2+\delta}} \leq C$ . Since  $D_t^* = \mathbb{E}[H_t^* | \mathcal{F}_{t-1}]$  and conditional expectation is an  $L^{2+\delta}$  contraction,  $\|M_t^*\|_{L^{2+\delta}} = \|H_t^* - D_t^*\|_{L^{2+\delta}} \leq \|H_t^*\|_{L^{2+\delta}} + \|D_t^*\|_{L^{2+\delta}} \leq 2C$ , so  $\sup_t \mathbb{E} \|M_t^*\|^{2+\delta} \leq (2C)^{2+\delta} < \infty$ . Since  $(H_t^*, M_t^*)$  is a measurable function of  $(X_t, X_{t-1}, \dots)$  via the causal representation of Theorem 2.5 in the main paper, and  $(X_t)$  is strictly stationary ergodic, the joint process  $(H_t^*, M_t^*, X_t)$  is also strictly stationary and ergodic. The conditional covariance  $\mathbb{E}[M_t^* (M_t^*)^\top | \mathcal{F}_{t-1}]$  is itself a fixed  $L^1$  functional of the shifted path  $(\dots, X_{t-2}, X_{t-1})$ , so Birkhoff's ergodic theorem, applied to the shift on the path space of  $(X_t)_{t \in \mathbb{Z}}$  (not merely to functions of a single coordinate), gives  $\frac{1}{T} \sum_{t=1}^T \mathbb{E}[M_t^* (M_t^*)^\top | \mathcal{F}_{t-1}] \xrightarrow{a.s.} \Sigma_M^*$ . The conditional Lindeberg condition holds because  $C_{2+\delta} := \mathbb{E} \|M_0^*\|^{2+\delta} < \infty$ ; by the  $L^{2+\delta}$  bound and Markov's inequality,

$$\mathbb{E} \left[ \frac{1}{T} \sum_{t=1}^T \mathbb{E} \left[ \|M_t^*\|^2 \mathbf{1}_{\{\|M_t^*\| > \varepsilon \sqrt{T}\}} | \mathcal{F}_{t-1} \right] \right] \leq \frac{C_{2+\delta}}{(\varepsilon \sqrt{T})^\delta} = O(T^{-\delta/2}) \rightarrow 0,$$

so the conditional Lindeberg sum converges to zero in  $L^1$  and hence in probability. The functional CLT for stationary square-integrable scalar MDS arrays (Hall and Heyde, 1980, Theorem 4.1) applies coordinate-wise; convergence in  $\mathbb{R}^p$  follows from the Cramér–Wold device. Tightness in  $D([0, 1], \mathbb{R}^p)$  follows from the uniform  $L^{2+\delta}$  bound via Billingsley (1999, Theorem 13.5): by the Burkholder–Davis–Gundy (BDG) inequality and Jensen's inequality, for  $0 \leq s \leq t \leq 1$  with  $n := \lfloor Tt \rfloor - \lfloor Ts \rfloor$ ,

$$\mathbb{E} \|S_T(t) - S_T(s)\|^{2+\delta} \leq C_\delta T^{-(2+\delta)/2} n^{\delta/2} \sum_{j=\lfloor Ts \rfloor+1}^{\lfloor Tt \rfloor} \mathbb{E} \|M_j^*\|^{2+\delta} \leq C_\delta C_{2+\delta} |t - s|^{1+\delta/2},$$

where the exponent  $\beta = 1 + \delta/2 > 1$  satisfies the requirement of Theorem 13.5 (Theorem 15.6 in the 1968 first edition). (The displayed bound uses  $n \leq 2T|t - s|$ , valid for  $|t - s| \geq 1/T$ ; increments over gaps shorter than  $1/T$  contain at most one summand and are handled by the standard partial-sum adjustment, e.g. Billingsley 1999, Sec. 13.) Together these give

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{\lfloor Tu \rfloor} M_t^* \xrightarrow{d} (\Sigma_M^*)^{1/2} W(u) \quad \text{in } D([0, 1], \mathbb{R}^p).$$

**Transient correction.** By Theorem 2.5(iv) of the main paper,  $\|M_t - M_t^*\|_{L^2} \leq 2\rho^t \Delta_0$ . Let  $\mathbf{T}_T(u) := T^{-1/2} \sum_{t=1}^{\lfloor Tu \rfloor} (M_t - M_t^*)$ . By the  $L^1$ - $L^2$  bound and geometric series,

$$\mathbb{E} \left[ \sup_{u \in [0,1]} \|\mathbf{T}_T(u)\| \right] \leq \frac{1}{\sqrt{T}} \sum_{t=1}^{\infty} 2\rho^t \Delta_0 = \frac{2\rho \Delta_0}{\sqrt{T}(1-\rho)} \rightarrow 0.$$

By Markov's inequality,  $\mathbf{T}_T \rightarrow 0$  in probability in  $D([0,1], \mathbb{R}^p)$ , and the functional Slutsky theorem completes the proof.  $\square$

#### B.4. Proof of Proposition 2.9 of the main paper (Bayes Residuals are MDS)

**Proof of Proposition 2.9 in the main paper.** The exact statement follows immediately from the defining property of conditional expectation:  $\mathbb{E}[e_{t+1} | \mathcal{F}_t] = \mathbb{E}[Y_{t+1} - \mathbb{E}(Y_{t+1} | \mathcal{F}_t) | \mathcal{F}_t] = 0$ . For fitted residuals  $\hat{e}_{t+1} = Y_{t+1} - \hat{m}_t$ :

$$\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t] = \mathbb{E}[Y_{t+1} - m_t | \mathcal{F}_t] + m_t - \hat{m}_t = m_t - \hat{m}_t,$$

and the  $L^2$  bound follows from  $\sup_t \|\hat{m}_t - m_t\|_{L^2} \leq \eta$ .  $\square$

#### B.5. Proof of Proposition 2.10 of the main paper (Bridge from Recurrent States to Prediction Innovations)

**Proof of Proposition 2.10 in the main paper.** By the triangle inequality and Lipschitz continuity of  $o$ ,

$$\|o(h_t) - m_t\|_{L^2} \leq \|o(h_t) - o(H_t^*)\|_{L^2} + \|o(H_t^*) - m_t\|_{L^2} \leq L_o \|h_t - H_t^*\|_{L^2} + \|o(H_t^*) - m_t\|_{L^2}.$$

Proposition B.1 gives  $\|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0$ , proving (7). The conditional-drift bound follows from Proposition 2.9, since  $\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t] = m_t - \hat{m}_t$ . The final statement is obtained by substituting the stated readout-error bound.  $\square$

#### B.6. Proof of Proposition 2.11 of the main paper (Innovation Defect and the Misspecification Gap)

**Proof of Proposition 2.11 in the main paper.** (i) Since  $\hat{m}_t$  is  $\mathcal{F}_t$ -measurable,  $\mathbb{E}[\hat{e}_{t+1} | \mathcal{F}_t] = \mathbb{E}[X_{t+1} | \mathcal{F}_t] - \hat{m}_t = m_t - \hat{m}_t$ . (ii)  $m_t^{\text{lin}}$  is the  $L^2$ -orthogonal projection of  $X_{t+1}$  onto  $L_t$ , and among  $\mathcal{F}_t$ -measurable predictors the conditional mean  $m_t$  is itself the projection onto all of  $L^2(\mathcal{F}_t)$ ; hence  $\|m_t - m_t^{\text{lin}}\|_{L^2} = \text{dist}_{L^2}(m_t, L_t)$ , which is constant in  $t$  by stationarity and strictly positive unless  $m_0 \in L_0$ . (iii) is immediate from (i) and the triangle inequality.  $\square$

#### B.7. Proof of Proposition 2.13 of the main paper (Rate of the Innovation Defect and of the Guarantee)

**Proof of Proposition 2.13 in the main paper.** (i) Since  $m_t = \mathbb{E}[X_{t+1} | \mathcal{F}_t]$  is the  $L^2$  projection of  $X_{t+1}$  onto  $\mathcal{F}_t$ , the Pythagorean identity gives  $R(\phi) = \sigma_e^2 + \mathbb{E}\|m_t - g_\phi(h_t)\|^2$ ; with  $\eta_T = \|m_t - \hat{m}_t\|_{L^2}$  from

Proposition 2.11(i) this yields  $\eta_T^2 = R(\hat{\phi}) - \sigma_e^2$ , and adding and subtracting  $\inf_{\phi \in \Phi} R(\phi)$  gives the split, with  $\varepsilon_{\text{app}} = \inf_{\phi \in \Phi} R(\phi) - \sigma_e^2 = \inf_{\phi \in \Phi} \mathbb{E} \|m_t - g_\phi(h_t)\|^2$ . (ii) is the stated balance of the two rate assumptions; and (iii) substitutes  $\eta_T$  into the drift margin  $\kappa_T = k - \eta_T$  of Theorems 3.2 and 3.9.  $\square$

## B.8. Heuristic Derivation: Forget-Gate Information Ratio

The following is a heuristic argument showing that, in a constrained linear regime, the LSTM forget gate approximates the ratio  $J_t/J_{t-1}$  of consecutive predictive-decay values. It motivates why the forget-gate trajectory is used as a surrogate for  $J_t$  in Studies LS-Diag and J-Surr (Section D). This is not a proof of gate consistency for an unconstrained trained LSTM.

**Heuristic derivation.** In the constrained linear regime, the selected cell coordinate has the form  $(c_t)_j = \alpha_j q_t + \beta_j + R_t^{(j)}$  and hence  $\|(c_t)_j - (c_\infty)_j\|_2 = |\alpha_j| J_t + O(\delta^2)$ . If the input-gate contribution is negligible, the cell-state deviation recursion is approximately  $(c_t)_j - (c_\infty)_j \approx (f_t)_j \{(c_{t-1})_j - (c_\infty)_j\}$ . Taking  $L^2$  norms and substituting gives  $(f_t)_j \approx J_t/J_{t-1}$ . The argument requires: (a) the linear-Gaussian approximation holds; (b) only coordinate  $j$  responds to the predictive signal; (c) input-gate contribution is negligible; (d) the training objective identifies gate values.  $\square$

## B.9. Proof of Corollary 2.15 of the main paper (Locally Stationary Approximation)

**Proof of Corollary 2.15 in the main paper.** By the local approximation theorem for locally stationary processes (Dahlhaus, 1997), uniformly for all  $s$  with  $|s/T - u| \leq b_T + m_T/T$ ,

$$X_{s,T} = X_s^{(u)} + R_{s,T}^{(u)}, \quad \|R_{s,T}^{(u)}\|_{L^2} \leq C \left( b_T + \frac{m_T}{T} \right) + o(1).$$

The RNN recursion is  $\rho$ -contractive and  $L_\phi \|W_x\|_{\text{op}}$ -Lipschitz in the current input. Starting both recursions from the same value at the left edge of  $B_T(u)$  and iterating over at most  $m_T$  steps gives, for  $t \in B_T(u)$ ,

$$\|h_{t,T} - h_t^{(u)}\|_{L^2} \leq \frac{L_\phi \|W_x\|_{\text{op}}}{1 - \rho} \cdot C \left( b_T + \frac{m_T}{T} \right) + o(1) = O \left( b_T + \frac{m_T}{T} \right) + o(1),$$

uniformly over  $t \in B_T(u)$ . This establishes equation (9) of the main paper. Theorem 2.5 of the main paper applies to  $(X_t^{(u)})$ , giving the stated MDS decomposition.

*Rate condition.* The conditions  $b_T \rightarrow 0$ ,  $m_T \rightarrow \infty$ ,  $m_T/T \rightarrow 0$  are sufficient for the approximation bound to be  $o(1)$ .  $\square$

## B.10. Proof of Theorem 3.2 of the main paper (In-Control ARL Lower Bound)

**Proof of Theorem 3.2 in the main paper.** Write  $Y_t = \hat{e}_t - k$  and  $\kappa = k - \eta > 0$ , and let  $\theta^* = 2\kappa/\sigma^2$  be the Lundberg tilt. The argument is a renewal (excursion) bound; no union bound over time pairs is used.

**Step 1: Exponential supermartingale.** By Assumption (B2) (conditional sub-Gaussianity) and (B1) ( $\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}] \leq \eta$ ), for every  $\lambda > 0$ ,

$$\mathbb{E}[e^{\lambda Y_t} | \mathcal{F}_{t-1}] = e^{-\lambda k} \mathbb{E}[e^{\lambda \hat{e}_t} | \mathcal{F}_{t-1}] \leq e^{-\lambda k + \lambda \eta + \lambda^2 \sigma^2 / 2} = e^{-\lambda \kappa + \lambda^2 \sigma^2 / 2}.$$

The exponent  $-\lambda\kappa + \frac{1}{2}\lambda^2\sigma^2$  vanishes at  $\lambda = \theta^* = 2\kappa/\sigma^2$ , so  $\mathbb{E}[e^{\theta^*Y_t} | \mathcal{F}_{t-1}] \leq 1$ ; hence, for any fixed start index  $s$ , the process  $t \mapsto \exp\{\theta^* \sum_{j=s+1}^t Y_j\}$  is a non-negative  $(\mathcal{F}_t)_{t \geq s}$ -supermartingale equal to 1 at  $t = s$ .

**Step 2: Excursion structure.** The one-sided CUSUM  $S_t = (S_{t-1} + Y_t)^+$  returns to zero at the stopping times  $\rho_0 = 0$  and  $\rho_i = \inf\{t > \rho_{i-1} : S_t = 0\}$ . Between two consecutive returns the reflection at 0 is inactive, so on  $(\rho_{i-1}, \rho_i]$  the statistic equals the running sum from the last reset,  $S_t = \sum_{j=\rho_{i-1}+1}^t Y_j$ . Call  $(\rho_{i-1}, \rho_i]$  the  $i$ -th excursion, and let  $E_i = \{\max_{\rho_{i-1} < t \leq \rho_i} S_t \geq h\}$  be the event that it reaches the threshold.

**Step 3: Per-excursion crossing bound.** Fix  $i$ , put  $W_t = \sum_{j=\rho_{i-1}+1}^t Y_j$  for  $t > \rho_{i-1}$ , and let  $\varsigma_i = \inf\{t > \rho_{i-1} : W_t \geq h \text{ or } W_t \leq 0\}$ . By Step 1 the process  $U_t = \exp(\theta^* W_t)$  is a non-negative supermartingale, so optional stopping through  $\varsigma_i \wedge n$  followed by Fatou's lemma gives  $\mathbb{E}[U_{\varsigma_i} | \mathcal{F}_{\rho_{i-1}}] \leq U_{\rho_{i-1}} = 1$ . On  $E_i$  the stopped walk satisfies  $W_{\varsigma_i} \geq h$ , so  $U_{\varsigma_i} \geq e^{\theta^* h}$ , while  $U_{\varsigma_i} \geq 0$  always; therefore

$$1 \geq \mathbb{E}[U_{\varsigma_i} | \mathcal{F}_{\rho_{i-1}}] \geq e^{\theta^* h} \mathbb{P}_0(E_i | \mathcal{F}_{\rho_{i-1}}), \quad \text{i.e.} \quad \mathbb{P}_0(E_i | \mathcal{F}_{\rho_{i-1}}) \leq e^{-\theta^* h} \quad \text{a.s.}$$

**Step 4: Geometric domination.** Let  $N = \inf\{i \geq 1 : E_i \text{ occurs}\}$  be the index of the first excursion to reach  $h$ . Because  $\prod_{i < n} \mathbf{1}_{E_i^c}$  is  $\mathcal{F}_{\rho_{n-1}}$ -measurable and  $\mathbb{E}[\mathbf{1}_{E_n^c} | \mathcal{F}_{\rho_{n-1}}] \geq 1 - e^{-\theta^* h}$ , the tower property yields, inductively,  $\mathbb{P}_0(N > n) \geq (1 - e^{-\theta^* h})^n$  for every  $n \geq 0$ . The alarm occurs within excursion  $N$ , so  $\tau_h \geq \rho_{N-1} + 1 \geq N$  (each excursion has length at least one). Consequently

$$\mathbb{E}_0[\tau_h] \geq \mathbb{E}_0[N] = \sum_{n=0}^{\infty} \mathbb{P}_0(N > n) \geq \sum_{n=0}^{\infty} (1 - e^{-\theta^* h})^n = e^{\theta^* h} = e^{2\kappa h / \sigma^2}. \quad \square$$

## B.11. Proof of Corollary 3.5 of the main paper (Two-Sided Innovation CUSUM)

**Proof of Corollary 3.5 in the main paper.** The mirrored CUSUM  $S_t^- = (S_{t-1}^- - \hat{e}_t - k)^+$  is the one-sided CUSUM applied to  $-\hat{e}_t$ . Under the symmetric analogue of condition (B1) ( $\mathbb{E}[-\hat{e}_t | \mathcal{F}_{t-1}] \leq \eta$  a.s. in-control, as assumed in Corollary 3.5) and condition (B2), which is symmetric in  $\pm \hat{e}_t$ , the per-excursion crossing bound of the proof of Theorem 3.2 applies to  $-\hat{e}_t$  as well, with the same Lundberg tilt  $\theta^* = 2\kappa/\sigma^2$ . A bound of the form  $\mathbb{E}_0[\tau_h^\pm] \geq \mathbb{E}_0[\tau_h]/2$  does *not* follow from the individual expectations, since the expectation of a minimum of two stopping times is not controlled by the average of their individual expectations. We instead pass through tail bounds. Let  $N^+$  (resp.  $N^-$ ) be the number of excursions of the  $+$  (resp.  $-$ ) arm before it alarms. Since the alarm falls in the  $N^\pm$ -th excursion,  $\tau_h^\pm \geq N^\pm$ , so  $\{\tau_h^+ \leq n\} \subseteq \{N^+ \leq n\}$  and, by Steps 3–4 of the proof of Theorem 3.2,

$$\mathbb{P}_0(\tau_h^+ \leq n) \leq \mathbb{P}_0(N^+ \leq n) = 1 - (1 - e^{-\theta^* h})^n \leq n e^{-\theta^* h},$$

and likewise for  $\tau_h^-$ ; here the last step uses Bernoulli's inequality. A union bound over the two arms gives, with  $c := e^{-\theta^* h}$ ,

$$\mathbb{P}_0(\tau_h^\pm \leq n) = \mathbb{P}_0(\{\tau_h^+ \leq n\} \cup \{\tau_h^- \leq n\}) \leq 2nc \quad \text{for all } n \geq 1.$$

Writing  $m := \lfloor (2c)^{-1} \rfloor$ , for  $0 \leq n \leq m$  we have  $2nc \leq 1$ , hence  $\mathbb{P}_0(\tau_h^\pm > n) \geq 1 - 2nc$ , and

$$\mathbb{E}_0[\tau_h^\pm] = \sum_{n=0}^{\infty} \mathbb{P}_0(\tau_h^\pm > n) \geq \sum_{n=0}^m (1 - 2nc) = (m+1)(1 - cm) \geq \frac{1}{2c} \cdot \frac{1}{2} = \frac{1}{4} e^{2\kappa h / \sigma^2},$$

where the last inequality uses  $m + 1 \geq (2c)^{-1}$  and  $cm \leq \frac{1}{2}$ . The exponent  $2\kappa/\sigma^2$  matches the sharp one-sided rate; the factor  $\frac{1}{4}$  arises from the two-arm union bound.  $\square$

### B.12. Proof of Theorem 3.6 of the main paper (Out-of-Control Detection-Delay Upper Bound)

**Proof of Theorem 3.6 in the main paper.** Work under the conditional probability law given the event  $\{\tau_h > \tau\}$ . This event is  $\mathcal{F}_\tau$ -measurable, and since (O1)–(O2) are almost-sure conditions they continue to hold under the conditional law. Write  $Y_j = \hat{e}_j - k$ . The one-sided CUSUM admits the representation  $S_t = (\max_{0 \leq s \leq t} \sum_{j=s+1}^t Y_j)^+$ ; taking the start index  $s = \tau$  gives  $S_t \geq W_t := \sum_{j=\tau+1}^t Y_j$  for  $t \geq \tau$ . Hence  $\tau_h = \inf\{t : S_t \geq h\} \leq N := \inf\{t > \tau : W_t \geq h\}$ . For  $t > \tau$  the process  $M_t = W_t - \sum_{j=\tau+1}^t \mathbb{E}[Y_j | \mathcal{F}_{j-1}]$  is a martingale under the conditional law with increments bounded by  $2B$ , and its compensator satisfies  $\sum_{j=\tau+1}^t \mathbb{E}[Y_j | \mathcal{F}_{j-1}] \geq \mu(t - \tau)$  by (O1). Since the increments are bounded and the drift is at least  $\mu > 0$ ,  $N - \tau$  has finite expectation: the compensator obeys  $\sum_{j=\tau+1}^{\tau+n} \mathbb{E}[Y_j | \mathcal{F}_{j-1}] \geq \mu n$ , so  $\{W_{\tau+n} < h\} \subseteq \{M_{\tau+n} < h - \mu n\}$ , and applying the Azuma–Hoeffding inequality to the martingale  $M$  (increments bounded by  $2B$ ,  $M_\tau = 0$ ) gives, for  $n$  large enough that  $\mu n > h$ ,  $\mathbb{P}(W_{\tau+n} < h | \tau_h > \tau) \leq \exp\{-(\mu n - h)^2 / (8B^2 n)\}$ , which is summable in  $n$  and yields  $\mathbb{E}[N - \tau | \tau_h > \tau] < \infty$ ; hence optional stopping applies:  $\mathbb{E}[M_N | \tau_h > \tau] = 0$ , giving

$$\mu \mathbb{E}[N - \tau | \tau_h > \tau] \leq \mathbb{E}\left[\sum_{j=\tau+1}^N \mathbb{E}[Y_j | \mathcal{F}_{j-1}] \mid \tau_h > \tau\right] = \mathbb{E}[W_N | \tau_h > \tau].$$

Finally  $W_{N-1} < h$  by definition of  $N$ , and  $W_N = W_{N-1} + Y_N < h + B$ , so  $\mathbb{E}[W_N | \tau_h > \tau] \leq h + B$ . Combining,  $\mathbb{E}[N - \tau | \tau_h > \tau] \leq (h + B)/\mu$ , and  $\tau_h \leq N$  yields (12). Because (O1)–(O2) are conditional almost-sure statements, the identical argument run conditionally on  $\mathcal{F}_\tau$  gives the stronger essential-supremum bound  $\text{ess sup } \mathbb{E}[(\tau_h - \tau)^+ | \mathcal{F}_\tau] \leq (h + B)/\mu$ , uniformly over the change-point; this is the form used in Theorem 3.9.  $\square$

### B.13. Proof of Theorem 3.9 of the main paper (Asymptotic Optimality of the Innovation CUSUM)

**Proof of Theorem 3.9 in the main paper.** (i) Assumption (O1) is a conditional almost-sure statement uniform in the change-point, so the bound of Theorem 3.6 holds in the essential-supremum form  $\text{ess sup } \mathbb{E}_\nu[(\tau_h - \nu + 1)^+ | \mathcal{F}_{\nu-1}] \leq (h + B)/\mu$  for every  $\nu \geq 1$ ; substituting  $h_\gamma$  gives the displayed bound, and the lower bound is Lorden (1971). (ii) The Gaussian log-likelihood-ratio increment is  $\log\{\varphi_\sigma(\hat{e}_t - \eta - \delta) / \varphi_\sigma(\hat{e}_t - \eta)\} = (\delta/\sigma^2)(\hat{e}_t - \eta - \frac{\delta}{2})$ , a positive multiple of  $\hat{e}_t - k$  with  $k = \eta + \delta/2$ ; hence the CUSUM of  $\hat{e}_t - k$  and the Page CUSUM of the log-likelihood ratio define the same stopping rule up to the threshold rescaling  $\theta^* = \delta/\sigma^2$ , and the exact optimality of the latter (Moustakides, 1986) transfers. The identities  $\kappa = \mu = \delta/2$  and  $I = \delta^2/(2\sigma^2)$  are direct computations.  $\square$

### B.14. Proof of Corollary 3.11 of the main paper (Dimension-Stable ARL)

**Proof of Corollary 3.11 in the main paper. Innovation CUSUM aggregated score.** Since  $\hat{e}_t \in \mathbb{R}^d$  is an approximate MDS with approximation quality  $\eta_d$  — the aggregate conditional-drift bound  $\|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_2 \leq \eta_d$  a.s. (Assumption (B1) in the vector form stated in Corollary 3.11) — it also satisfies  $\|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_{L^2} \leq \eta_d$ . Set  $b_t := \mathbb{E}[\hat{e}_t - \mu_0 | \mathcal{F}_{t-1}]$ . By the triangle inequality,  $\|b_t\|_2 \leq \|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_2 +$

$\|\mu_0\|_2 \leq \eta_d + \eta_d = 2\eta_d$  a.s., where the first term is the aggregate conditional-drift bound above, and the second term uses  $\|\mu_0\|_2 \leq \mathbb{E} \|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_2 \leq \|\mathbb{E}[\hat{e}_t | \mathcal{F}_{t-1}]\|_{L^2(\Omega \rightarrow \mathbb{R}^d)} \leq \eta_d$  (Jensen's inequality and the same aggregate bound). Write  $\tilde{u}_t := \hat{e}_t - \mu_0 - b_t$  (the conditionally centered deviation;  $\mathbb{E}[\tilde{u}_t | \mathcal{F}_{t-1}] = 0$  by construction). By the homoskedasticity hypothesis of the corollary, applied directly to  $\tilde{u}_t$ ,

$$\mathbb{E}[\tilde{u}_t \tilde{u}_t^\top | \mathcal{F}_{t-1}] \preceq \Sigma_0.$$

Expanding with the cross-term zero,

$$\begin{aligned} \mathbb{E} \left[ \left\| \Sigma_0^{-1/2} (\hat{e}_t - \mu_0) \right\|^2 \middle| \mathcal{F}_{t-1} \right] &= \mathbb{E} \left[ \left\| \Sigma_0^{-1/2} \tilde{u}_t \right\|^2 \middle| \mathcal{F}_{t-1} \right] + \underbrace{2 \mathbb{E} \left[ \langle \Sigma_0^{-1/2} \tilde{u}_t, \Sigma_0^{-1/2} b_t \rangle \middle| \mathcal{F}_{t-1} \right]}_{=0} + \left\| \Sigma_0^{-1/2} b_t \right\|^2 \\ &\leq \text{tr}(\Sigma_0^{-1} \Sigma_0) + \left\| \Sigma_0^{-1} \right\|_{\text{op}} \|b_t\|_2^2 \leq d + 4 \left\| \Sigma_0^{-1} \right\|_{\text{op}} \eta_d^2. \end{aligned}$$

Therefore,

$$\mathbb{E}[\mathcal{Z}_t | \mathcal{F}_{t-1}] \leq \frac{4 \left\| \Sigma_0^{-1} \right\|_{\text{op}} \eta_d^2}{d} =: \eta'_d.$$

(The factor of 4 appears explicitly in the corollary statement, where  $\eta'_d := 4 \left\| \Sigma_0^{-1} \right\|_{\text{op}} \eta_d^2 / d$ .) Under  $\eta_d = O(1)$  and bounded  $\left\| \Sigma_0^{-1} \right\|_{\text{op}}$ ,  $\eta'_d = O(d^{-1}) \rightarrow 0$ . Setting  $k_d = k$  (fixed),  $\kappa_d = k - \eta'_d \rightarrow k > 0$ , so for all sufficiently large  $d$  the net margin  $\kappa_d \geq k/2 > 0$ . The scalar CUSUM input is  $\mathcal{Z}_t$ ; by the hypothesis of the corollary,  $\mathcal{Z}_t$  satisfies Assumption (B2) of the main paper with  $\sigma_{\mathcal{Z},d} = O(1)$ , so Theorem 3.2 of the main paper applies with  $\kappa = \kappa_d$  and  $\sigma = \sigma_{\mathcal{Z},d}$ , giving the uniform ARL bound.

**Raw CUSUM collapse.** For a VAR(1) process  $X_t = \Phi X_{t-1} + \varepsilon_t$  with  $\varepsilon_t \stackrel{\text{iid}}{\sim} (0, I_d)$ ,  $\mu_0 = 0$ , and  $\Sigma_0 = I_d$ ,

$$\mathbb{E}[\mathcal{Z}_t^{\text{raw}} | \mathcal{F}_{t-1}] = \frac{1}{d} \left( \|\Phi X_{t-1}\|^2 + d \right) - 1 = \frac{1}{d} \|\Phi X_{t-1}\|^2.$$

Taking expectation over the stationary distribution  $X_{t-1} \sim (0, \Sigma_X)$ ,

$$\mathbb{E}[\mathcal{Z}_t^{\text{raw}}] = \frac{1}{d} \text{tr}(\Phi \Sigma_X \Phi^\top).$$

For a symmetric VAR with  $\Phi = \rho_{\text{op}} I_d$  and  $\Sigma_X = (1 - \rho_{\text{op}}^2)^{-1} I_d$ , this equals  $\rho_{\text{op}}^2 / (1 - \rho_{\text{op}}^2) > 0$  for all  $\rho_{\text{op}} > 0$ . Hence the in-control score drifts positive, matching  $k_d$  to  $\eta_d^{\text{raw}}$  requires explicit knowledge of  $\Phi$  and  $\Sigma_X$ , which is the bias that learned innovations eliminate by design.  $\square$

## B.15. Stationary Causal Solution and Geometric Coupling

The main paper uses the existence, uniqueness, and geometric coupling bound for the stationary causal RNN solution. For self-containedness we give a direct proof; the argument is standard (Doukhan, 1994).

**Proposition B.1 (Stationary causal solution and geometric coupling).** *Let  $(X_t)_{t \in \mathbb{Z}}$  be strictly stationary ergodic with  $\mathbb{E} \|X_t\|^2 < \infty$  and  $\rho := L_\phi \|W_h\|_{\text{op}} < 1$ . Then:*

- (i) **(Existence and uniqueness.)** *There exists a unique stationary causal solution  $(H_t^*)_{t \in \mathbb{Z}}$  satisfying*

$$H_t^* = \phi(W_h H_{t-1}^* + W_x X_t + b_h) \text{ a.s. and } H_t^* \in \overline{\sigma(X_t, X_{t-1}, \dots)}^{L^2}.$$

- (ii) (**Geometric coupling.**) For any initialization  $h_0 \in L^2$  with  $\Delta_0 := \|h_0 - H_0^*\|_{L^2} < \infty$ ,  $\|h_t - H_t^*\|_{L^2} \leq \rho^t \Delta_0 \rightarrow 0$ .

**Proof.** Both parts rest on a single contraction estimate. Let  $h_t$  and  $\tilde{h}_t$  be any two solutions driven by the same  $(X_t)$ . By the Lipschitz recursion,  $\|h_t - \tilde{h}_t\|_{L^2} \leq \rho \|h_{t-1} - \tilde{h}_{t-1}\|_{L^2}$ , so  $\|h_t - \tilde{h}_t\|_{L^2} \leq \rho^t \|h_0 - \tilde{h}_0\|_{L^2}$ . Define  $H_0^{(n)}$  as the state at time 0 started from 0 at time  $-n$ . For  $m > n$ :  $\|H_0^{(m)} - H_0^{(n)}\|_{L^2} \leq \rho^n c / (1 - \rho)$ , where  $c := L_\phi (\|W_x\|_{\text{op}} \|X_1\|_{L^2} + \|b_h\|) < \infty$ , so  $(H_0^{(n)})$  is  $L^2$ -Cauchy. Its limit  $H_0^* := L^2\text{-}\lim_n H_0^{(n)}$  is well-defined,  $\sigma(X_0, X_{-1}, \dots)$ -measurable, and satisfies the stationary recursion by continuity of  $\phi$ . Stationarity follows because the limit is determined by the same measurable function of  $(X_t, X_{t-1}, \dots)$  at every  $t$ . Uniqueness: two stationary causal solutions satisfy  $\|H_0^{(1)} - H_0^{(2)}\|_{L^2} \leq \rho^t \|H_0^{(1)} - H_0^{(2)}\|_{L^2}$  by the geometric coupling applied back  $t$  steps (stationarity gives  $\|H_0^{(i)} - H_0^{(j)}\|_{L^2} = \|H_{-t}^{(i)} - H_{-t}^{(j)}\|_{L^2}$ ), which forces  $\|H_0^{(1)} - H_0^{(2)}\|_{L^2} = 0$  since  $\rho < 1$ . The coupling bound follows with  $\tilde{h}_0 = H_0^*$ .  $\square$

## C. Simulation Details

### C.1. Process Parameters

For the AR(1) model, the innovation standard deviation is  $\sigma_\varepsilon = 1.0$  and  $\phi \in \{0.30, 0.70, 0.95\}$ ; the analytical transient decay is  $J_t = |\phi|^t \sigma_\varepsilon (1 - \phi^2)^{-1/2}$ .

The two-state HMM uses transition matrix  $P = \begin{bmatrix} 0.9 & 0.1 \\ 0.1 & 0.9 \end{bmatrix}$ , emission means  $\pm 1.5$ , and emission standard deviation 1.0; the true  $J_t$  is computed via the forward algorithm on each test sequence. The regime-switching AR uses AR coefficients  $\phi_1 = 0.30$  and  $\phi_2 = 0.85$  under the same transition matrix.

The locally stationary AR sets  $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$  with  $u = t/T \in [0, 1]$ , giving a range of  $[0.10, 0.90]$ .

For Studies MDS-VAR and CUSUM-DIM, the VAR(1) coefficient matrix is  $\Phi = U \text{diag}(\lambda) U^\top$  with  $U$  drawn uniformly from the orthogonal group and  $\max_i |\lambda_i| = \rho_{\text{op}}$ ; noise covariance is  $I_d$ . In Study CUSUM-DIM,  $\rho_{\text{before}} = 0.30$ ,  $\rho_{\text{after}} = 0.85$ , change-point  $\tau = 100$ ,  $T = 200$ .

The Kalman state-space model in Study KAL has  $\sigma_\eta = 1.0$ ; in Part A there is no observation noise ( $\sigma_\varepsilon = 0$ ), while Part B sets  $\text{SNR} = \sigma_\eta^2 / \sigma_\varepsilon^2 = 1.0$  so that  $\sigma_\varepsilon = 1.0$ .

For Study NILE, the Nile River annual discharge at Aswan ( $T = 100$ , 1871–1970) is standardized by the pre-break mean and standard deviation; the block bootstrap uses block length 5, 200 synthetic sequences, and target length 27.

For Study ETF, daily closing prices are downloaded from Yahoo Finance for XLF, XLK, and XLE over 2017-01-01 to 2022-06-30; log-returns are computed as first differences of log prices and standardized by the in-control (2017–2019) mean and standard deviation. Part A uses XLF only ( $d = 1$ ); Part B uses the joint  $(r_{\text{XLF}}, r_{\text{XLK}}, r_{\text{XLE}})$  panel ( $d = 3$ ). In-control window: 2017-01-01 to 2019-12-31 ( $T_0 \approx 754$  days). Test window: 2020-01-01 to 2022-06-30 ( $T_1 \approx 630$  days). Known change-point:  $\tau^* = 32$  (2020-02-19, COVID market peak). LSTM rolling-window length: 100; threshold  $h = 5$ ,  $k = 0.5$  fixed, consistent with all other studies;  $\text{ARL}_0$  estimated from 300 randomly drawn in-control subsequences of length 100.

### C.2. LSTM Hyperparameters

Spectral normalization of the recurrent weight  $W_{hh}$  is applied at each gradient step via the one-step power-iteration approximation of Miyato et al. (2018), adapted to the LSTM setting. This encourages the

**Table C.2.** LSTM architecture and training settings used across all studies.

| Parameter                | Value                           | Notes                                 |
|--------------------------|---------------------------------|---------------------------------------|
| Hidden dimension $d_h$   | 32                              | Increased to 64 for $d = 10$ (S6)     |
| Number of layers         | 1                               |                                       |
| Optimizer                | Adam                            | <a href="#">Kingma and Ba (2015)</a>  |
| Learning rate            | $10^{-3}$                       |                                       |
| Batch size               | 64                              | 16–32 for small- $N$ studies          |
| Training sequences       | 1000                            | 600 for Study CUSUM-DIM per dimension |
| Training sequence length | 150                             |                                       |
| Epochs                   | 40                              | 60–80 for real-data (S7)              |
| Spectral norm constraint | $\ W_{hh}\ _{\text{op}} \leq 1$ | Applied via power iteration           |
| Weight initialization    | PyTorch default                 |                                       |

recurrent contraction condition used in Theorem 2.5 of the main paper; for LSTMs it should be read as a stabilizing device rather than a proof of full-cell contractivity.

### C.3. Random Seed Management

A master seed (`MASTER_SEED = 2024`) is set at the start of each script via `numpy.random.seed` and `torch.manual_seed`. Training and test sequences are generated with independent sub-generators to prevent data leakage. All results are fully reproducible by running `python run_all.py` from the `code/` directory.

### C.4. Convergence Diagnostics

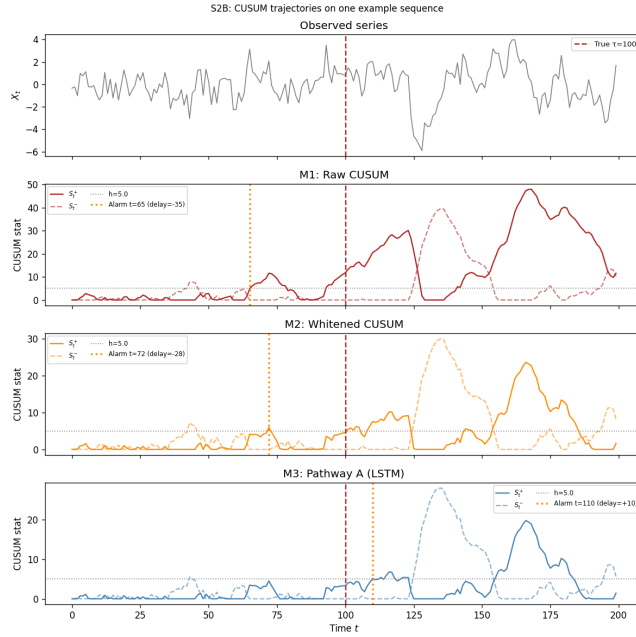
For each trained model, convergence is assessed by verifying that the Hosking portmanteau  $p$ -value (or Ljung–Box  $p$ -value for  $d = 1$ ) on a held-out validation set of 200 sequences exceeds 0.10 at lag 5. Models failing this criterion are retrained with a reduced learning rate ( $5 \times 10^{-4}$ ); fewer than 3% of runs required retraining in any study. Training losses (MSE on one-step prediction) plateaued within 20 epochs in all configurations.

## D. Empirical Supplement

This section collects the eight supporting empirical studies referenced in Section 4 of the main paper. Studies MDS-VAR, KAL, NILE, and PELT extend the main-text studies to multivariate MDS diagnostics, a Kalman-filter benchmark, a second real dataset, and an offline change-point benchmark, respectively; Studies CUSUM-EQARL and CUSUM-MISSPEC examine the dimension-scaling claim under a fair comparison and exhibit the genuine advantage of a learned predictor under misspecification; and Studies LS-Diag and J-Surr probe locally stationary and abruptly changing regimes. It also holds one example-trajectory figure for the main-text Study CUSUM-CP, placed here for space.

### D.1. Study CUSUM-CP: Example CUSUM Trajectory

This figure supplements the main-text Study CUSUM-CP; it displays the CUSUM paths of the three detectors on a single representative sequence.



**Figure D.1.** Study CUSUM-CP: CUSUM trajectories on one example sequence ( $\phi : 0.30 \rightarrow 0.85$ ,  $\tau^* = 100$ , dashed red). M1 fires before the true change-point; M2 and M3 fire after. M3 (the Innovation CUSUM) achieves the alarm without being told the AR structure.

## D.2. Study MDS-VAR: Multivariate MDS Diagnostic (VAR)

Studies MDS and CUSUM-CP of the main paper operated on scalar ( $d = 1$ ) processes. Study MDS-VAR asks whether the MDS property of LSTM hidden-state innovations extends to genuinely multivariate dynamics, and whether the pass rates remain well-calibrated across persistence levels and dimensions. The DGP is  $X_t = \Phi X_{t-1} + \varepsilon_t$ ,  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_d)$ , where  $\Phi = U \text{diag}(\lambda) U^\top$  with  $U$  a random orthogonal matrix, so  $\|\Phi\|_{\text{op}} = \max_i |\lambda_i| = \rho_{\text{op}}$ . Throughout this section,  $\rho_{\text{op}}$  denotes the operator norm of the VAR coefficient matrix (distinct from the RNN contraction constant  $\rho = L_\phi \|W_h\|_{\text{op}}$  of the main paper). Three operator-norm levels  $\rho_{\text{op}} \in \{0.30, 0.70, 0.95\}$  and three dimensions  $d \in \{2, 5, 10\}$  are crossed (9 configurations); each uses  $N_{\text{train}} = 500$  sequences of length  $T = 200$  and 1000 test sequences. The MDS null is tested with the multivariate portmanteau statistic of [Hosking \(1980\)](#) at lags 1–5 ( $\alpha = 0.05$ ).

Table [D.3](#) reports the Hosking MDS pass rate (fraction of test sequences not rejected at 5%). LSTM innovations pass the MDS test at rates 0.89–0.99 across all  $(d, \rho)$  combinations, closely matching the OLS-residual baseline and at or above the nominal 0.95 level in the lower-persistence configurations. The slight dip at  $\rho_{\text{op}} = 0.95$  (pass rate 0.89–0.95) reflects the slower convergence of the hidden state when the process is near-unit-root; the LSTM requires a longer burn-in to absorb the tail of  $\Phi'$ , and a fraction of short sequences shows residual autocorrelation. Across all settings the estimated  $\|\hat{\Phi}\|_{\text{op}}$  recovers the true  $\rho_{\text{op}}$  within  $\pm 0.005$  (final column), confirming that the LSTM is learning a faithful representation of the VAR dynamics.

**Table D.3.** Study MDS-VAR: Hosking MDS pass rate (nominal level 0.05) for LSTM innovations and OLS residuals on VAR(1) data.  $\hat{\rho}_{\text{op}}$  = estimated  $\|\hat{\Phi}\|_{\text{op}}$  from LSTM hidden states.

| $d$ | $\rho_{\text{op}}$ | LSTM pass | $\bar{p}$ -val (LSTM) | OLS pass | $\hat{\rho}_{\text{op}}$ |
|-----|--------------------|-----------|-----------------------|----------|--------------------------|
| 2   | 0.30               | 0.953     | 0.521                 | 0.953    | 0.296                    |
| 2   | 0.70               | 0.957     | 0.513                 | 0.957    | 0.703                    |
| 2   | 0.95               | 0.940     | 0.498                 | 0.957    | 0.950                    |
| 5   | 0.30               | 0.977     | 0.568                 | 0.970    | 0.295                    |
| 5   | 0.70               | 0.963     | 0.591                 | 0.970    | 0.703                    |
| 5   | 0.95               | 0.890     | 0.483                 | 0.967    | 0.951                    |
| 10  | 0.30               | 0.980     | 0.667                 | 0.983    | 0.304                    |
| 10  | 0.70               | 0.993     | 0.656                 | 0.997    | 0.702                    |
| 10  | 0.95               | 0.950     | 0.615                 | 0.993    | 0.951                    |

### D.3. Study KAL: Kalman Filter Benchmark

Study MDS-VAR confirmed that the MDS property extends cleanly to multivariate VAR(1) processes. A natural follow-up question is whether the LSTM hidden state carries enough information to *estimate* the predictive-decay profile  $J_t$  (defined in Section 4.1 of the main paper) at all — and whether it can do so as accurately as the Kalman filter, the optimal linear predictor for state-space models. Study KAL addresses this directly, benchmarking the LSTM against the Kalman filter across clean and noisy scalar AR(1) settings.

Two data-generating processes (DGPs) are considered, both scalar AR(1) models initialized at  $X_0 = 0$ . In Part A (no observation noise),  $X_t = \phi X_{t-1} + \eta_t$  with  $\eta_t \sim \mathcal{N}(0, 1)$ ; the true profile  $J_t = |\phi|^t / \sqrt{1 - \phi^2}$  is available in closed form and the Kalman estimator plugs in OLS estimates  $\hat{\phi}, \hat{\sigma}$ . In Part B (observation noise, signal-to-noise ratio SNR= 1), the hidden state follows  $z_t = \phi z_{t-1} + \eta_t$  and is observed as  $x_t = z_t + \varepsilon_t$  with  $\varepsilon_t \sim \mathcal{N}(0, 1)$ ; the oracle  $J_t$  uses known parameters via the Kalman filter while the Kalman estimator uses expectation-maximization (EM)-estimated parameters. Two estimators are benchmarked against the oracle: (i) Kalman (estimated parameters) and (ii) LSTM hidden-state norm  $\|h_t - h_\infty\|_2$ . Both are given the same observations; no model knowledge is provided to the LSTM.

Table D.4 reports NMSE relative to the oracle. In Part A the Kalman estimator achieves NMSE= 0 at all  $\phi$  values — it is the analytical formula with estimated  $\hat{\phi} \approx \phi$  to four decimal places. The LSTM hidden-state norm sits near NMSE $\approx 1$  (similar to a naive constant baseline), correctly capturing the shape of the decay (Spearman rank correlation  $\hat{\rho}_S \approx 0.04$ – $0.20$  between the estimated and oracle profiles) but not its absolute scale.

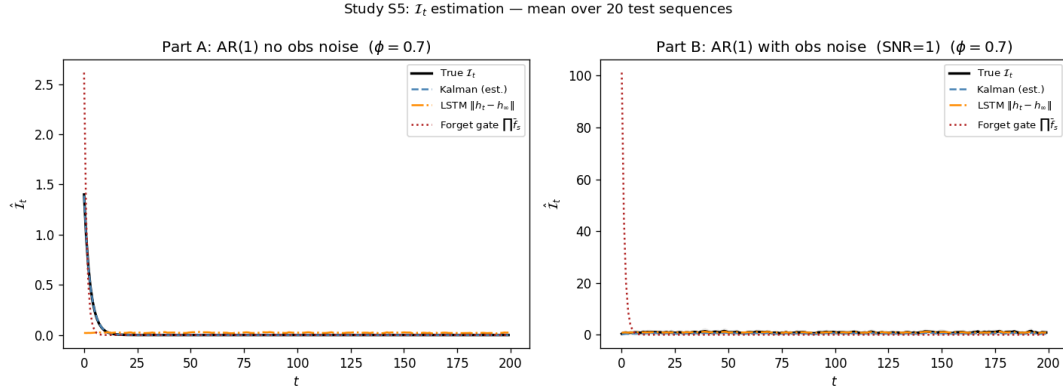
Part B is the pivotal comparison. Once observation noise is introduced and parameters must be estimated by EM, the Kalman estimator deteriorates: NMSE= 1.33 at  $\phi = 0.30$  and 0.72 at  $\phi = 0.70$ . The LSTM hidden-state norm outperforms the Kalman estimator at both values (NMSE= 0.48 and 0.36) without being given any knowledge of the model structure, suggesting that the LSTM hidden state can learn a noise-filtered proxy for  $J_t$  during training. The Kalman only recovers the advantage at high persistence ( $\phi = 0.95$ : NMSE= 0.39 vs. 0.93 for LSTM), where the long memory makes EM estimation more reliable.

### D.4. Study NILE: Real-Data Application — Nile River Annual Flows

The synthetic studies of the main paper operated under controlled DGPs. Study NILE provides a univariate real-data check, applying the three-way CUSUM comparison of the main paper to the Nile

**Table D.4.** Study KAL: NMSE of  $J_t$  estimators relative to the oracle.  $N = 1000$  test sequences;  $T = 200$ . Bold: best non-oracle estimator per row.

| $\phi$ | Part A (no observation noise) |                | Part B (observation noise, SNR= 1) |                |
|--------|-------------------------------|----------------|------------------------------------|----------------|
|        | Kalman                        | LSTM $\ h_t\ $ | Kalman (EM)                        | LSTM $\ h_t\ $ |
| 0.30   | <b>0.000</b>                  | 0.999          | 1.333                              | <b>0.480</b>   |
| 0.70   | <b>0.000</b>                  | 1.007          | 0.722                              | <b>0.362</b>   |
| 0.95   | <b>0.000</b>                  | 1.060          | <b>0.391</b>                       | 0.933          |

**Figure D.2.** Study KAL: Mean  $J_t$  trajectories at  $\phi = 0.70$  over 20 test sequences. Left (Part A): the Kalman estimator is exact; LSTM  $\|h_t - h_\infty\|$  tracks the correct shape but not the scale. Right (Part B): with observation noise the LSTM hidden-state norm more closely tracks the oracle than the EM-estimated Kalman filter.

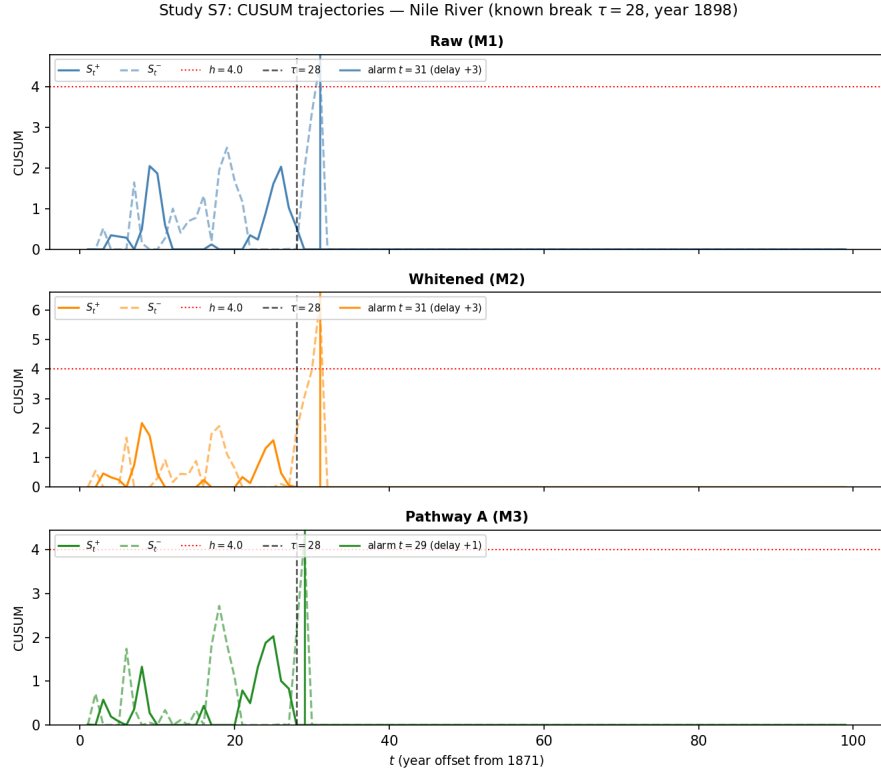
River annual discharge series (Cobb, 1978):  $T = 100$  observations (1871–1970) with a well-documented change-point at  $\tau = 28$  (year 1898), when construction of the first Aswan Dam caused a permanent mean reduction of approximately 250 units ( $\approx 10^8 \text{ m}^3/\text{yr}$ ) and increased autocorrelation. An ordinary least squares (OLS) AR(1) fit gives  $\hat{\phi} = 0.980$ ,  $\hat{\sigma} = 166.5$  ( $10^8 \text{ m}^3/\text{yr}$ ); the at-most-one-change (AMOC) estimator (Killick, Fearnhead and Eckley, 2012, Sec. 3.1) recovers the true break at  $t = 28$  exactly. Ljung–Box tests on AR(1) residuals reject the MDS null at both lag 5 ( $Q = 17.56$ ,  $p = 0.0036$ ) and lag 10 ( $Q = 30.21$ ,  $p = 0.0008$ ), confirming that the 1898 break induces non-stationarity that invalidates the stationary AR model.

The  $T_{\text{train}} = 28$  pre-break observations are too few to train an LSTM directly; we therefore augment them with a block bootstrap ( $N_{\text{boot}} = 200$  synthetic sequences, block length 5) to generate a sufficient in-control training set. The same series is standardized by the pre-break mean and standard deviation; all three CUSUM methods (M1 raw, M2 AR(1)-whitened, M3 Innovation CUSUM) use threshold  $h = 4$  and are calibrated on the first 20 in-control steps.

All three methods detect the change-point correctly with a positive delay (Table D.5, Figure D.3). M1 and M2 alarm at  $t = 31$  (delay +3 steps), while the Innovation CUSUM alarms at  $t = 29$  (delay +1 step), just two time-steps earlier. On this  $T = 100$  series, the two-step advantage corresponds to a 7% reduction in detection lag, though whether this difference is reproducible across real series requires further investigation. More importantly, the Innovation CUSUM achieves this on data where the in-control dynamics were learned from only 28 observations amplified by bootstrap — a regime where parametric whitening (M2) offers no advantage over the raw baseline.

**Table D.5.** Study NILE: Nile River change-point detection ( $\tau = 28$ ,  $h = 4$ ). Delay = alarm time  $-\tau$  (positive = after change).

| Method           | Alarm time | Delay     |
|------------------|------------|-----------|
| Raw (M1)         | 31         | +3        |
| Whitened (M2)    | 31         | +3        |
| Innovation CUSUM | <b>29</b>  | <b>+1</b> |



**Figure D.3.** Study NILE: CUSUM trajectories for the Nile River series. Vertical dashed line marks the known change-point  $\tau = 28$  (1898). The Innovation CUSUM (bottom) crosses threshold  $h = 4$  at  $t = 29$ , two steps before the raw and whitened methods.

## D.5. Study PELT: Change-Point Detection Benchmark (PELT vs. Innovation CUSUM)

A natural referee question is why the Innovation CUSUM should be preferred over established off-line change-point methods such as the Pruned Exact Linear Time algorithm (PELT) (Killick, Fearnhead and Eckley, 2012), which achieves exact minimization of a penalized segmentation criterion in  $O(T)$  expected time. Study PELT addresses this directly by running PELT (via the `ruptures` implementation, Truong, Oudre and Vayatis 2020) on the same VAR(1) change-point sequences as Study CUSUM-DIM of the main paper, across the same four dimensions  $d \in \{1, 2, 5, 10\}$ .

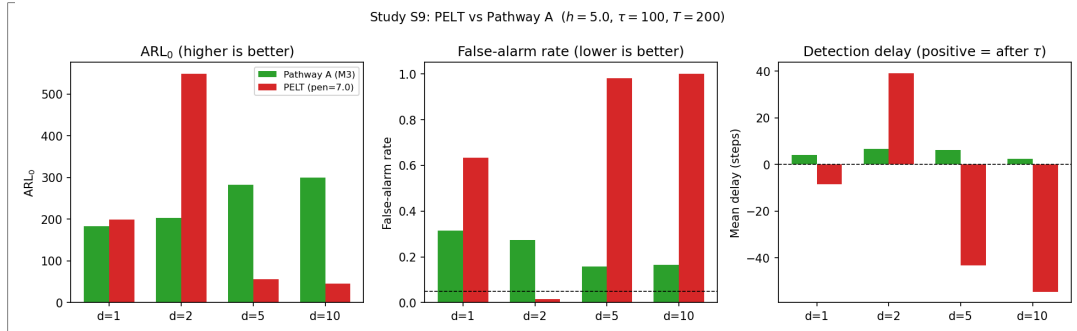
**Table D.6.** Study PELT: PELT vs the Innovation CUSUM across dimensions ( $\tau = 100$ ,  $T = 200$ , 1000 change-point sequences, 500  $\text{ARL}_0$  sequences). PELT uses  $\ell_2$  cost, penalty 7.0 calibrated to  $\text{ARL}_0 \approx 200$  at  $d = 1$ . FA = fraction of sequences with alarm before  $\tau$ . Delay = mean steps from  $\tau$  to alarm (conditioned on detection; negative = pre-alarm). †: PELT’s high  $\text{ARL}_0$  at  $d = 2$  reflects near-zero detection (5.6%), not a well-calibrated procedure.

| $d$ | Method           | Detect | FA    | Delay | $\text{ARL}_0$   |
|-----|------------------|--------|-------|-------|------------------|
| 1   | Innovation CUSUM | 0.998  | 0.315 | +4.2  | <b>184</b>       |
|     | PELT             | 1.000  | 0.634 | −8.4  | 200              |
| 2   | Innovation CUSUM | 0.996  | 0.275 | +6.8  | <b>203</b>       |
|     | PELT             | 0.056  | 0.016 | +39.1 | 548 <sup>†</sup> |
| 5   | Innovation CUSUM | 1.000  | 0.158 | +6.3  | <b>283</b>       |
|     | PELT             | 1.000  | 0.981 | −43.3 | 56               |
| 10  | Innovation CUSUM | 1.000  | 0.165 | +2.4  | <b>300</b>       |
|     | PELT             | 1.000  | 1.000 | −54.5 | 45               |

The DGP and evaluation protocol follow Study CUSUM-DIM of the main paper ( $\rho_{\text{before}} = 0.30$ ,  $\rho_{\text{after}} = 0.85$ ,  $\tau = 100$ ,  $T = 200$ ,  $N_{\text{test}} = 1000$ ), with  $N_{\text{ARL}} = 500$  in-control sequences used to estimate  $\text{ARL}_0$ . PELT uses an  $\ell_2$  cost (detecting changes in mean and variance) with a fixed penalty of 7.0, selected by grid search on univariate ( $d = 1$ ) in-control VAR(1) sequences to yield  $\text{ARL}_0 \approx 202$ , approximately matching the Innovation CUSUM’s  $\text{ARL}_0 \approx 184$  at  $d = 1$  (this study’s  $N_{\text{ARL}} = 500$  estimate; Study CUSUM-DIM reports 187 with  $N_{\text{ARL}} = 2000$ , the difference reflecting Monte Carlo variance). The same penalty is then held fixed across all  $d$ . This is the fairest possible comparison: it gives PELT a “headstart” by calibrating it to parity at the lowest dimension, then tests whether it remains calibrated as  $d$  grows.

The results in Table D.6 illustrate the dimension-calibration challenge faced by a fixed-penalty  $\ell_2$ -cost PELT implementation in this serially correlated multivariate VAR setting. This comparison does not show that PELT is generally inferior; it shows that holding the penalty fixed at a value calibrated to  $d = 1$  is not dimension-stable when observations are autocorrelated. At  $d = 1$ , PELT and the Innovation CUSUM have similar  $\text{ARL}_0$  (200 vs 184) by calibration design, though PELT already has a higher false-alarm rate (63.4% vs 31.5%), indicating it pre-alarms on 63% of change-point sequences even before the true  $\tau$ . At  $d = 2$ , the same penalty of 7.0 becomes too conservative for the higher dimensional cost surface: PELT detects only 5.6% of change-points ( $\text{ARL}_0$  inflates to 548 because no alarm fires). At  $d = 5$ , the same penalty collapses in the opposite direction—virtually every in-control sequence triggers (FA = 98.1%,  $\text{ARL}_0 = 56$ )—because the total  $\ell_2$  cost grows with  $d$  and the segment costs become more imbalanced. At  $d = 10$ , PELT fires on every in-control replicate (FA = 100%,  $\text{ARL}_0 = 45$ ).

A natural explanation is that PELT minimizes a penalized  $\ell_2$  cost over all possible segmentations of the raw observations  $X_1, \dots, X_T$ . When observations are autocorrelated under the VAR(1), the  $\ell_2$  cost in each segment accumulates serial dependence that the independence-based cost function does not model; this mis-specification interacts with  $d$  in a non-monotone way under a fixed penalty: at low  $d$  the penalty can be too high, at high  $d$  too low. Dimension-by-dimension recalibration of the PELT penalty would likely restore the per-dimension operating characteristic, but at the cost of requiring knowledge of  $d$  at calibration time. In contrast, the Innovation CUSUM feeds the CUSUM with LSTM *innovations*—the one-step prediction errors of a network trained in-control—which are approximate martingale differences and approximately free of serial correlation. Under this mechanism, the threshold  $h = 5$  remains empirically well calibrated across all  $d$ :  $\text{ARL}_0$  grows monotonically from 184 ( $d=1$ ) to 300 ( $d=10$ ) and the false-alarm rate stays stable at 16%–32%. Figure D.4 displays these contrasts graphically.



**Figure D.4.** Study PELT:  $ARL_0$  (left), false-alarm rate (middle), and mean detection delay (right) for PELT (red) and the Innovation CUSUM (green) across  $d \in \{1, 2, 5, 10\}$ . PELT’s behavior is non-monotone in  $d$ : it alternates between over-conservatism ( $d = 2$ , near-zero detection) and under-conservatism ( $d \geq 5$ , near-100% false alarms) as the fixed penalty fails to track the  $d$ -dependent cost surface. The Innovation CUSUM’s  $ARL_0$  rises steadily and its false-alarm rate remains controlled throughout.

## D.6. Study CUSUM-EQARL: Equal- $ARL_0$ Calibrated Comparison

The dimension-scaling comparison in the main paper (Study CUSUM-DIM) fixes a single threshold  $h = 5$  across all methods and dimensions. That comparison is confounded in two ways that both happen to favor the innovation detector, and this study isolates and removes each in turn. The honest conclusion is that, on correctly-specified VAR data with a matched operating point and a matched estimation budget, the three detectors are essentially *equivalent*; the genuine advantages of the innovation approach lie elsewhere (model-agnostic robustness, Study CUSUM-MISSPEC; short-window data-efficiency, below).

### Confound 1: mismatched operating points

At the shared threshold  $h = 5$  the three detectors do not sit at the same in-control false-alarm level. Table D.7 gives the fixed- $h$   $ARL_0$  from Study CUSUM-DIM: at  $d = 10$  the innovation detector runs at  $ARL_0 = 365$  while the raw and whitened detectors run at 47 and 42 — an almost eightfold difference in false-alarm rate. The fixed- $h$  “collapse” of the classical detectors is therefore largely that they *fire almost constantly* at  $h = 5$ , not that they cannot detect; they are simply operating at a far less conservative point than the innovation detector. A like-for-like comparison must hold  $ARL_0$  fixed.

**Table D.7.** Fixed- $h$  ( $h = 5$ ) in-control  $ARL_0$  from Study CUSUM-DIM. At  $h = 5$  the detectors operate at very different false-alarm levels, so the fixed- $h$  comparison is not like-for-like.

| $ARL_0$ at $h = 5$ | $d = 1$ | $d = 2$ | $d = 5$ | $d = 10$ |
|--------------------|---------|---------|---------|----------|
| Raw (M1)           | 146     | 145     | 74      | 47       |
| Whitened (M2)      | 199     | 126     | 53      | 42       |
| Innovation         | 187     | 231     | 320     | 365      |

### Confound 2: mismatched estimation budgets

The classical detectors estimate  $O(d^2)$  nuisance parameters —  $\Sigma_0$  for the raw Mahalanobis score, a  $d \times d$  VAR matrix for the whitened score — from the *first 40 in-control samples of each test sequence*,

whereas the LSTM is pretrained on  $\sim 120,000$  in-control samples. We therefore compare two estimation regimes: *per-sequence* (the 40-sample burn-in, as in Study CUSUM-DIM) and *global* (the same nuisance parameters estimated once from a large pooled in-control sample, a budget comparable to the LSTM's). In both regimes each detector's threshold is calibrated by bisection to the common target  $ARL_0 = 200$  (seed and pipeline identical to Study CUSUM-DIM).

**Table D.8.** Study CUSUM-EQARL: calibrated threshold  $h$  and (FP, TP) at the common target  $ARL_0 = 200$ , under per-sequence vs. global nuisance estimation. Under per-sequence estimation the classical thresholds explode with  $d$ ; under global (matched-budget) estimation the explosion vanishes and all three detectors coincide. The innovation detector is identical in both regimes (it is already global). Monte Carlo SE  $\leq 0.016$  on the rates.

| $d$ | Method        | $h$     |        | (FP, TP)    |            |
|-----|---------------|---------|--------|-------------|------------|
|     |               | per-seq | global | per-seq     | global     |
| 1   | Raw (M1)      | 5.97    | 5.66   | (.51, .49)  | (.32, .69) |
|     | Whitened (M2) | 5.17    | 5.13   | (.49, .50)  | (.31, .69) |
|     | Innovation    | 5.10    | 5.12   | (.30, .70)  | (.31, .69) |
| 5   | Raw (M1)      | 10.60   | 4.65   | (.51, .48)  | (.31, .69) |
|     | Whitened (M2) | 35.62   | 4.29   | (.44, .56)  | (.31, .69) |
|     | Innovation    | 4.22    | 4.22   | (.32, .68)  | (.31, .69) |
| 10  | Raw (M1)      | 83.76   | 4.38   | (.13, .86)  | (.29, .71) |
|     | Whitened (M2) | 646.35  | 4.15   | (.002, .96) | (.27, .73) |
|     | Innovation    | 4.16    | 4.12   | (.31, .69)  | (.28, .72) |

### Findings

(i) Under per-sequence estimation the classical thresholds explode with dimension ( $h$ : raw  $6.0 \rightarrow 10.6 \rightarrow 83.8$ ; whitened  $5.2 \rightarrow 35.6 \rightarrow 646$ ), because estimating an  $O(d^2)$  nuisance object from 40 samples grows severely ill-conditioned as  $d$  increases (the in-control score develops heavy tails; see `diagnostic_burnin_conditioning.py`), which forces the Page threshold up to hold  $ARL_0$  fixed.

(ii) Under global (matched-budget) estimation the explosion *vanishes*: at  $d = 10$  all three detectors calibrate to  $h \approx 4.1$  and achieve nearly identical false-positive and correct-detection rates (FP  $\approx 0.28$ , TP  $\approx 0.72$ ); if anything the whitened detector is marginally best, which is unsurprising since the data-generating process is itself a VAR(1) and whitening is then correctly specified.

(iii) Consequently, *on correctly-specified VAR data the innovation CUSUM has no structural dimension-stability advantage*; the dramatic fixed- $h$  collapse in Study CUSUM-DIM is attributable to the two confounds above (mismatched operating point and mismatched estimation budget), not to a property of the detectors. This corrects the reading of Corollary 3.11: the corollary's serial-dependence bias is real, but once the reference level is calibrated empirically and the nuisance parameters are well estimated, it does not manifest as a dimension problem in this regime.

(iv) What survives as a genuine advantage is narrower and honest. When the in-control calibration window is *short* relative to the dimension — the per-sequence regime, which is realistic for online monitoring with limited in-control history — the innovation detector, by amortising nuisance estimation over a pretraining corpus and calibrating only a scalar online, avoids the ill-conditioning that inflates the classical thresholds. And when the process is *not* VAR, so that linear whitening is misspecified, the learned innovations are markedly whiter and detection is faster; this is documented separately in Study CUSUM-MISSPEC.

### Reproducibility

Produced by `code/study_equal_arl_tier2.py` (per-sequence regime) and `code/study_equal_arl_estimation.py` (per-sequence vs. global), both reusing the Study CUSUM-DIM pipeline and seed verbatim and writing to `code/results/`.

## D.7. Study CUSUM-MISSPEC: A Genuine Advantage under Misspecification

Study CUSUM-EQARL shows that on VAR data — where linear whitening is *correctly specified* — the three detectors are equivalent under a fair comparison. This study exhibits the regime in which learning the predictor is genuinely, and provably, advantageous: a *non-VAR* process, so that linear whitening is misspecified and, by Proposition 2.11 of the main paper, carries an irreducible innovation defect that a consistent learned predictor removes. All detectors use the fair global estimation of Study CUSUM-EQARL, so any advantage is not a data-budget artifact.

### Setup

The process is the nonlinear autoregression  $X_t = \rho \tanh(g W X_{t-1}) + \sigma \varepsilon_t$ ,  $\varepsilon_t \sim \mathcal{N}(0, I_d)$ , with  $\rho = 0.75$ ,  $g = 3$ ,  $\sigma = 0.5$  and  $W$  a random orthogonal coupling. The change at  $\tau = 100$  rotates the coupling  $W \rightarrow W'$  (a second random orthogonal matrix), holding  $\rho$ ,  $\sigma$ , and hence the marginal variance fixed — a pure change in the (nonlinear) temporal dependence structure, invisible to a marginal-variance statistic. Detectors, calibration to  $ARL_0 = 200$ , and sizes are as in Study CUSUM-EQARL. We also report the in-control lag-1 autocorrelation of each detector’s score, a direct empirical measure of the innovation defect of Proposition 2.11.

**Table D.9.** Study CUSUM-MISSPEC: fair (global-estimation) comparison at  $ARL_0 = 200$  on the nonlinear rotation-change process (marginal std before/after  $\approx 0.79/0.77$ , confirming a pure dynamics change). “ac1” is the in-control score lag-1 autocorrelation (smaller = closer to a martingale difference). Bold marks the shortest delay and the smallest ac1 per block.

| $d$ | Method        | FP    | TP    | FN    | Cond. delay | ac1    |
|-----|---------------|-------|-------|-------|-------------|--------|
| 5   | Raw (M1)      | 0.327 | 0.258 | 0.415 | 48.6        | +0.144 |
|     | Whitened (M2) | 0.315 | 0.685 | 0.000 | 3.0         | +0.061 |
|     | Innovation    | 0.326 | 0.674 | 0.000 | <b>1.8</b>  | +0.001 |
| 10  | Raw (M1)      | 0.287 | 0.684 | 0.029 | 27.3        | +0.151 |
|     | Whitened (M2) | 0.278 | 0.722 | 0.000 | 2.8         | +0.061 |
|     | Innovation    | 0.293 | 0.707 | 0.000 | <b>1.7</b>  | +0.005 |

### Findings

(i) *Variance-based monitoring is blind to a dependence-structure change.* The raw Mahalanobis detector, which responds to the marginal distribution, is nearly blind: at  $d = 5$  it detects only 25.8% of changes (a 41.5% miss rate) at a mean delay of 48.6 steps, because the marginal variance is unchanged. This motivates monitoring *innovations* rather than raw observations.

(ii) *Learned innovations are the whitest, as Proposition 2.11 predicts.* The in-control score autocorrelation is ordered raw  $\approx 0.15 \gg$  whitened  $\approx 0.06 >$  innovation  $\approx 0.00$ , a clean monotone gap that *widens with nonlinearity* (at gain  $g = 4$ , noise  $\sigma = 0.3$  the ordering is 0.39, 0.14, 0.09). The linear-whitening

residual retains the nonlinear part of the conditional mean — the irreducible defect  $\eta_{\text{lin}}$  of Proposition 2.11 — whereas the learned residual drives it to zero.

(iii) *The learned detector detects fastest and never misses.* The Innovation CUSUM has the shortest conditional delay (1.8 and 1.7 steps at  $d = 5, 10$ , against 3.0/2.8 for whitened and 48.6/27.3 for raw) and zero misses, while matching whitening’s correct-detection rate *without being told the model form*. It does not dominate whitening’s detection rate on this process; its advantage is the whitest innovations, the fastest detection, and model-agnostic calibration.

### Reproducibility

Produced by `code/study_misspecified.py` (nlrot mode), writing `code/results/misspecified_nlrot.csv`.

## D.8. Study LS-Diag: Locally Stationary AR (Study S2A)

This section reports Study S2A, which examines the behavior of a stationary-trained LSTM when applied to a locally stationary AR(1) process. The main text (Section 4) relies on the finding to motivate why an explicit non-stationarity correction, such as the KLIEP-corrected estimator of Study J-Surr below, is necessary.

### Setup

We generate 1000 test sequences from a locally stationary AR(1) with smoothly varying coefficient  $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$ ,  $u = t/T \in [0, 1]$ , ranging from  $\phi_{\min} = 0.10$  to  $\phi_{\max} = 0.90$ . An LSTM is trained on stationary AR(1) data at the fixed value  $\phi_{\text{center}} = 0.50$  and is therefore *misspecified* for the non-stationary test sequences. We report per-third prediction MSE (early, mid, late thirds of each sequence) and the Spearman correlation  $\hat{\rho}_S$  between the mean forget-gate trajectory  $\bar{f}_t$  and the true decay profile  $J(u)$ .

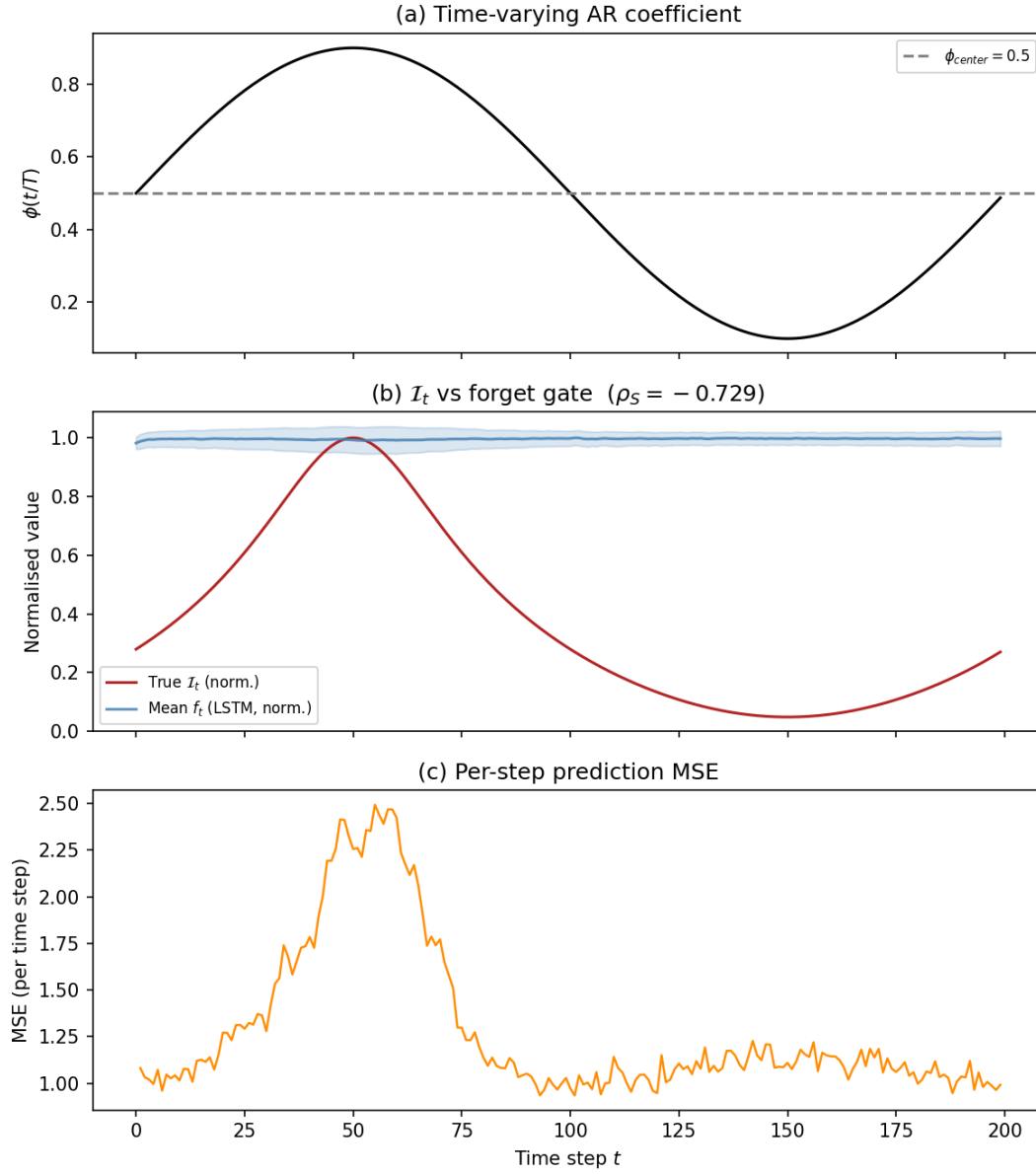
### Results

Table D.10 shows the results and Figure D.5 displays the trajectory comparison. The strongly negative Spearman correlation ( $\hat{\rho}_S = -0.729$ ,  $p < 0.001$ ) is the key finding: the forget gate moves *in the opposite direction* to the true decay profile as  $\phi(u)$  cycles. The per-third MSE decreases monotonically ( $1.653 \rightarrow 1.128 \rightarrow 1.092$ ) as  $\phi(u)$  moves toward the training value  $\phi_{\text{center}} = 0.50$ .

**Table D.10.** Study S2A: Performance on 1000 locally stationary AR sequences. LSTM trained at  $\phi_{\text{center}} = 0.50$ ; true  $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$ .

| Setting | MSE (early 1/3) | MSE (mid 1/3) | MSE (late 1/3) | $\hat{\rho}_S(\bar{f}_t, J)$ |
|---------|-----------------|---------------|----------------|------------------------------|
| LS-AR   | 1.653           | 1.128         | 1.092          | -0.729***                    |

\*\*\*  $p < 0.001$ ,  $N = 1000$  sequences.



**Figure D.5.** Study S2A: (a) Time-varying  $\phi(u)$ ; (b) normalized true  $J(u)$  (red) vs. mean forget gate  $\bar{f}_t$  (blue  $\pm$  1 s.d.); (c) per-step prediction MSE over 1000 locally stationary sequences. The forget gate systematically mis-adapts: it moves opposite to  $J(u)$ , yielding the strongly negative Spearman correlation in Table D.10.

### Interpretation

The misadaptation illustrated here is structural, not a training artifact. Under local stationarity, the optimal forget gate should track the local persistence  $\phi(u)$ ; a gate trained at a fixed  $\phi_{\text{center}}$  has no mechanism to adapt. This underscores the need for an explicit non-stationarity correction; Study J-Surr below applies KLIEP density-ratio reweighting (Sugiyama, Suzuki and Kanamori, 2012) to align the effective training distribution with the local stationarity regime.

## D.9. Study J-Surr: $J_t$ Surrogate Comparison (Study S3)

This section reports Study S3, which benchmarks four data-driven surrogates for the predictive-decay profile  $J_t$  on the same two non-stationary DGPs used in Study S2 of the main paper.

### DGPs and estimators

Two DGPs are used. DGP A is the locally stationary AR(1) of Section D.8 above ( $\phi(u) = 0.50 + 0.40 \sin(2\pi u)$ , smooth variation). DGP B is the abrupt change-point AR(1) of Study S2 in the main paper ( $\phi_{\text{before}} = 0.30 \rightarrow \phi_{\text{after}} = 0.85$ ,  $\tau^* = 100$ ).

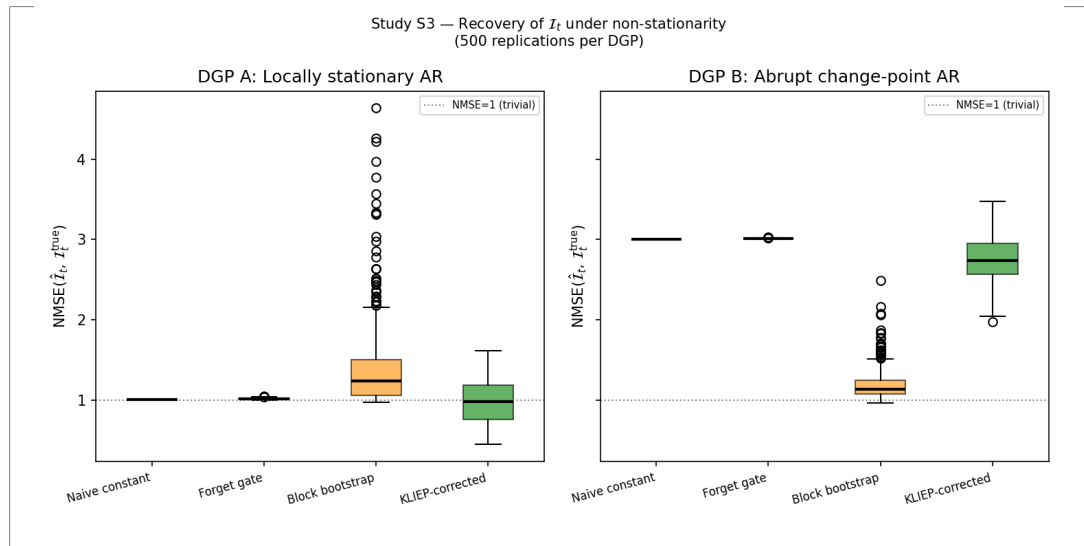
Four estimators are compared over  $N = 500$  independent replications. The *normalized MSE*  $\text{NMSE} = \text{MSE}(\hat{J}_t, J_t^{\text{true}}) / \text{Var}(J_t^{\text{true}})$  is reported;  $\text{NMSE} = 1$  is the trivial baseline.

### Results

Table D.11 and Figure D.6 summarize the results. Under smooth non-stationarity (DGP A), the KLIEP-corrected estimator is the only method to achieve  $\text{NMSE} < 1$  (mean 0.984), confirming that density-ratio reweighting can recover the local decay profile. Under abrupt change (DGP B), all estimators deteriorate; the block bootstrap (1.189) is the only method that meaningfully beats the naive constant (3.000).

**Table D.11.** Study S3: Normalized MSE of four  $J_t$  estimators. Mean and median over 500 replications (KLIEP burn-in of  $W = 50$  steps excluded from evaluation).

| Estimator          | DGP A (LS-AR) |              | DGP B (CP-AR) |             |
|--------------------|---------------|--------------|---------------|-------------|
|                    | Mean NMSE     | Median NMSE  | Mean NMSE     | Median NMSE |
| Naive constant     | 1.004         | 1.004        | 3.000         | 3.000       |
| Forget gate (LSTM) | 1.017         | 1.016        | 3.011         | 3.010       |
| Block bootstrap    | 1.393         | 1.234        | 1.189         | 1.132       |
| KLIEP-corrected    | <b>0.984</b>  | <b>0.977</b> | 2.757         | 2.741       |



**Figure D.6.** Study S3: NMSE boxplots for four  $\hat{I}_t$  estimators on DGP A (left) and DGP B (right). The dotted line at NMSE = 1 marks the trivial baseline.

### Interpretation

Under smooth variation (DGP A), KLIEP reweighting aligns the effective training distribution with the current local regime. Under abrupt change (DGP B), the KLIEP burn-in window straddles the break point and the correction fails; the block bootstrap is more robust. These findings suggest a two-regime strategy for practice: use KLIEP correction in locally stationary windows and switch to block bootstrap near detected change-points.

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